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A MODIFIED EULER-MARUYAMA METHOD TO SIMULATE A GENERAL ONE-DIMENSIONAL STICKY DIFFUSION

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Abstract. A sticky diffusion is a process that can stick to and detach from a lower-dimensional boundary. A challenge in simulating such a process is in capturing the change in dimension in a dynamically consistent way. We introduce a numerical algorithm to simulate a one-dimensional sticky diffusion with general drift and diffusion, which sticks to and detaches from a point. Our method is a simple modification of the standard Euler-Maruyama scheme, which chooses with some probability between a reflected Euler-Maruyama update and a jump to the sticky point. We show how to choose this probability to be consistent with the generator of the desired dynamics, and we prove that our scheme converges weakly to a sticky diffusion with order 1.

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1. Introduction

Sticky diffusions are diffusion processes that can spend a nonzero amount of time on a lower-dimensional boundary. The simplest example is sticky Brownian motion, which behaves as Brownian motion away from the origin but is slowed at the origin so that it spends a positive amount of time there. More generally, a sticky diffusion follows a stochastic differential equation (SDE) in the ambient space and may follow different dynamics on the boundary.

The mathematical study of sticky diffusions began with Feller's classification of the possible boundary behaviours of one-dimensional diffusion processes [16, 17], followed by developments in probability and PDE theory [24, 26, 42, 47, 50, 51].

After being somewhat neglected for several decades, sticky diffusions have regained attention because of their wide range of applications. In finance, they model variables that linger near thresholds, including interest rates near zero [1, 27, 32, 39]. In physics and chemistry, they describe particles with short-ranged adhesive interactions that may form transient bonds [4, 22, 35, 36, 41, 46]. Biological applications include molecular capture and release on cell surfaces [9, 19], pathogen load with a sticky state at zero [10, 44], and models for biomass and rare-species abundance [38]. Sticky diffusions also arise in queues and storage processes [20], transport and communication networks [5], and coupling arguments used to establish convergence rates for sampling algorithms and SDEs [12, 13, 23].

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Simulating a sticky diffusion is challenging because the boundary carries a singular measure. Standard SDE methods such as Euler–Maruyama (EM) use random spatial steps and therefore do not naturally place the numerical process exactly on the boundary [28]. One approach approximates stickiness through a strong, short-ranged force near the boundary [7, 15, 43], but resolving the resulting stiffness requires timesteps much smaller than the timescales of interest. Continuous-time Markov-chain approximations instead take random steps in time on a predefined set of spatial points [7, 8, 33, 34]. Although effective for low-dimensional problems with flat boundaries, these methods scale poorly with dimension and differ substantially from standard diffusion-simulation workflows, making them difficult to incorporate into existing simulation methodologies [2, 18, 48].

A recent reflected-EM method adjusts the timestep when a boundary is crossed [45]. It applies to sticky diffusions obtained through a time change of a reflected diffusion process, but does not readily extend to arbitrary sticky diffusions that cannot be constructed in this way. Moreover, the numerical process never moves exactly to the boundary, so the method cannot directly exploit established methods for simulating constrained dynamics [3, 6, 11, 29, 49].

Our contribution. In this paper we make an important step toward simulating arbitrary sticky diffusions by proposing a modified Euler–Maruyama method for a one-dimensional sticky reflecting diffusion process. The numerical scheme can move exactly to and from a sticky boundary point. It operates in discrete time, is only a small modification of the standard reflected EM scheme, and is conceptually simple to understand and implement.

The method works as follows. When the process is sufficiently far from its sticky point, which we take to be at $x = 0$, the algorithm performs a reflected EM update. When the process is close to 0, it performs this update with probability $\lambda(x)$, depending on its current position x, and moves exactly to 0 with probability $1 - \lambda(x)$. The function λ is chosen so that the one-step transition operator of the discrete scheme approximates the transition operator of the continuous dynamics over a timestep h.

We prove that the method converges weakly with order $O(h)$. Away from the boundary, the local error is $O(h^2)$ over $O(h^{-1})$ timesteps, giving an $O(h)$ contribution to the global error. Near the boundary, the local error is $O(h^{3/2})$, while the expected number of such timesteps is $O(h^{-1/2})$, producing the same global order.

Overview of the paper. Section 2 reviews one-dimensional sticky diffusions and their associated forward and backward equations. Section 3 develops the method first for sticky Brownian motion (SBM), including the derivation of λ, numerical experiments, and the convergence analysis. We begin with SBM because the construction is clearest in this setting and the general analysis reuses several estimates established there. Section 4 extends the method and convergence analysis to a general one-dimensional sticky diffusion. The main results and supporting lemmas are presented in that section, while technically involved proofs using standard techniques are collected in the Supplement.

2. A brief introduction to one-dimensional sticky diffusion processes

In this section we give an overview of the key equations characterizing one-dimensional sticky diffusions. Our overview will be mainly formal, with the goal of providing accessible ways of writing down the equations characterizing such diffusions.

2.1. **Stochastic differential equation**

A sticky diffusion process is the weak solution to a particular stochastic differential equation (SDE). Though not the main tool we use to characterize such processes, we present the SDE here for completeness.

We consider a process which, on domain $D = (0, \infty)$, behaves as a diffusion process with drift $b(x)$ and diffusion $\sigma(x)$. At the boundary, $\partial D = \{0\}$, the process is delayed before it is reflected, in such a way that it cumulatively spends a finite, nonzero amount of time there. The process can be constructed as the solution to the following system of stochastic differential equations [25]:

$$dX_t = b(X_t)1_D(X_t)dt + \sigma(X_t)1_D(X_t)dW_t + dL_t, \tag{2.1a}$$

$$\frac{\sigma^2(X_t)}{2}1_{\partial D}(X_t)dt = \kappa dL_t. \tag{2.1b}$$

Here $1_A(\cdot)$ is the indicator function for set A, W_t is a one-dimensional Brownian motion, and L_t is the local time process, a nondecreasing process which increases only when X_t is on the boundary, as

$$L_t = \int_0^t 1_{\partial D}(X_s)dL_s.$$

The constant $\kappa > 0$ characterizes how "sticky" the origin is. Larger κ implies the origin is stickier, while $\kappa = 0$ is not at all sticky and corresponds to a purely reflected process.

The existence and uniqueness of weak solutions to (2.1) is shown in [25, Theorem 7.2], where a sufficient condition is shown to be[1] that σ, b are bounded and Lipschitz continuous on $(0,\infty)$ with $\sigma(0)^2 > 0$.

2.2. **Backward equation**

The key object we will work with to show weak convergence of our numerical scheme is the backward equation associated with (2.1). We start by recalling the generator for (2.1) is the differential operator $\mathcal{L}$ given by

$$\mathcal{L}f(x) = b(x)\partial_x f + a(x)\partial_{xx} f. \tag{2.2}$$

As usual $a(x) = \frac{1}{2}\sigma^2(x)$. The sticky boundary implies a restriction on the domain of the generator, formulated as the sticky boundary condition [25]

$$a\partial_x f = \kappa\mathcal{L}f \qquad \text{at } x = 0. \tag{2.3}$$

The backward equation is thus characterized by the following.

Proposition 2.1. *Let $\phi : [0,\infty) \to \mathbb{R}$ be a bounded function, let $T > 0$, and suppose the backward equation*

$$\partial_s u + \mathcal{L}u = 0, \qquad (x \geq 0) \tag{2.4a}$$

$$a\partial_x u = \kappa\mathcal{L}u \qquad (x = 0) \tag{2.4b}$$

$$u(T, x) = \phi(x). \tag{2.4c}$$

has a solution $u : [0,T] \times [0,\infty) \to \mathbb{R}$ which is such that $u, \partial_t u, \partial_x u, \partial_{xx} u$ are continuous and bounded on $[0,T]\times[0,\infty)$, and $\sigma\partial_x u$ is bounded on that same domain. Then u has the representation $u(t,x) = \mathbb{E}[\phi(X_T)|X_t = x]$, where $(X_t)_{t\geq 0}$ is a weak solution to (2.1)*.*

Proof. Let u solve (2.4). Using Itô's formula, rewriting $1_D\mathcal{L}u = \mathcal{L}u - 1_{\partial_D}\mathcal{L}u$, and using (2.1b), we evaluate:

$$\begin{aligned}
&u(T, X_T) \\
&\quad = u(t, X_t) + \int_t^T (\partial_s u + 1_D\mathcal{L}u)(s, X_s)ds + \int_t^T (\sigma 1_D \partial_x u)(s, X_s)dW_s + \int_t^T \partial_x u(s, X_s)dL_s \\
&\quad = u(t, X_t) + \int_t^T (\partial_s u + \mathcal{L}u)(s, X_s)ds + \int_t^T (\sigma 1_D \partial_x u)(s, X_s)dW_s + \int_t^T 1_{\partial D}(-\mathcal{L}u + \frac{a}{\kappa}\partial_x u)(s, X_s))ds.
\end{aligned}$$

The deterministic integrals exist because of the regularity assumptions on u and its derivatives, and the stochastic integral is a martingale because of the regularity assumption on $\sigma\partial u$.

[1]This reference shows existence and uniqueness for a general d-dimensional sticky diffusion on a half space but we specialize the equation and remarks here to a one-dimensional process.

The first integrand is identically zero, by (2.4a). The last integrand is identically zero, by the sticky boundary condition (2.4b). Taking $\mathbb{E}[\cdot|X_t = x]$ shows that $\mathbb{E}[u(X_T,T)|X_t = x] = u(x,t)$. Then we apply the terminal condition (2.4c) to obtain the result. □

2.3. Stationary distribution

One way to see the role of parameter κ in determining the stickiness of the process (2.1) is to compute the stationary distribution. We will show (formally) that a locally stationary distribution is

$$\pi(x) = \tilde{\pi}(x)(1 + \kappa\delta(x)), \tag{2.5}$$

where $\tilde{\pi}$ is a locally stationary distribution for a Markov process with generator (2.2) assuming reflecting boundary conditions. That is, $\tilde{\pi}$ satisfies $\mathcal{L}^*\tilde{\pi} = 0$, where $\mathcal{L}^*$ is the formal adjoint of $\mathcal{L}$:

$$\mathcal{L}^*p = \partial_x(-b(x)p + \partial_x(a(x)p)). \tag{2.6}$$

Furthermore we have $-b\pi + \partial_x(a\pi)|_{x=0,\infty} = 0$ by the reflecting boundary condition on $\tilde{\pi}$. Hence, the κ is directly proportional to the atom of probability on the boundary.

We use a weak formulation of the usual condition $\mathcal{L}^*\pi = 0$ for a stationary distribution, which asks to show that $\langle \pi, \mathcal{L}f\rangle = 0$ for all f in the domain of the generator, where $\langle f, g\rangle = \int_0^\infty fgdx$ is the L_2 inner product on $\bar{D}$. We make use of an easy calculation:

$$p\mathcal{L}f = f\mathcal{L}^*p + \partial_x\Big(f\big(pb - \partial_x(pa)\big) + pa\partial_x f\Big). \tag{2.7}$$

Now we may calculate:

$$\begin{aligned}
\langle \pi, \mathcal{L}f\rangle &= \tilde{\pi}\kappa\mathcal{L}f\big|_{x=0} + \Big(\tilde{\pi}a\partial_x f + f(\tilde{\pi}b - \partial_x(\tilde{\pi}a)\Big)\Big|_{x=0}^{\infty} + \int_0^\infty f\mathcal{L}^*\tilde{\pi}dx \\
&= \tilde{\pi}(\kappa\mathcal{L}f - a\partial_x f) \\
&= 0.
\end{aligned}$$

The last two terms in the first line were zero because of the condition that $\tilde{\pi}$ was a locally stationary distribution for $\mathcal{L}^*$. The final terms vanished by the boundary condition (2.3).

2.4. Derivation via conservation of probability

We'll now consider an alternative formulation of the evolution equations for a sticky diffusion, in terms of the evolution of its probability measure. This formulation can be derived from simple considerations of conservation of probability. Consider a process that has probability density $p_1(x,t)$ for $x > 0$, and which has an atom of probability $p_0(t)$ at 0, so the total density of the process can be written as

$$\rho(x,t) = p_1(x,t)1(x > 0) + p_0(t)\delta(x).$$

Suppose p_1 evolves with the diffusion dynamics described by the operator $\mathcal{L}^*$ in (2.6):

$$\partial_t p_1 = \mathcal{L}^*p_1.$$

Let's write $j = bp_1 - \partial_x(ap_1)$ for the flux associated with $\mathcal{L}^*$, so that $\mathcal{L}^*p_1 = -\partial_x j$.

Probability is exchanged with the atom via flux from the exerior:

$$\partial_t p_0 = -j|_{x=0}.$$

One can verify this condition conserves the total probability $\int_0^\infty \rho dx$.

We need one additional condition to close this system and obtain a well-posed system of equations for p_0, p_1. One such condition is

$$p_0(t) = \kappa p_1(0,t)$$

for some constant $\kappa > 0$. This condition was shown in [7, 22] to be consistent with the stickiness near zero arising from the limit of a strong, short-ranged force pushing the diffusion process towards 0.

Putting this together and using that $\partial_t p_0 = \kappa \partial_t p_1|_{x=0} = \kappa \mathcal{L}^* p_1|_{x=0}$, shows the probability density is

$$\rho(x,t) = p_1(x,t)(1 + \kappa\delta(x)) \tag{2.8}$$

where p_1 solves

$$\partial_t p_1 = \mathcal{L}^* p_1 \quad (x \geq 0), \qquad \kappa\mathcal{L}^* p_1 = -j \quad (x = 0), \tag{2.9}$$

with $j = bp_1 - \partial_x(ap_1)$.

2.5. Equivalence of the forward and backward equations

Finally, let's show that the operator $\mathcal{L}^*$ in (2.9) with the given boundary condition, is the (formal) adjoint of $\mathcal{L}$ with boundary condition (2.3). We wish to show that

$$\langle f, \partial_t \rho \rangle = \langle \mathcal{L} f, \rho \rangle \tag{2.10}$$

for all test functions f satisfying (2.3), and for all ρ of the form (2.8) where p_1 is a test function (infinitely differentiable with compact support). This is a weak form of the usual condition $\langle f, \mathcal{L}^*\rho\rangle = \langle \mathcal{L} f, \rho\rangle$, and is more suited to handling δ-functions. We calculate

$$\langle f, \partial_t \rho \rangle = \int_0^\infty f \partial_t p_1 dx \quad + f\kappa\partial_t p_1\Big|_{x=0},$$

and, using (2.7),

$$\begin{aligned}\langle \mathcal{L} f, \rho \rangle &= \int_0^\infty f \mathcal{L}^* p_1 dx \quad - \Big(fj + p_1 a \partial_x f\Big)\Big|_{x=0} \quad + \kappa p_1 \mathcal{L} f\Big|_{x=0} \\ &= \int_0^\infty f\mathcal{L}^* p_1 dx - fj\Big|_{x=0} \quad ,\end{aligned}$$

by the sticky boundary condition (2.3). Thus, we can see that (2.10) holds, exactly when p_1 satisfies (2.9).

3. Sticky Brownian motion

In this section we present our algorithm for simulating a sticky Brownian motion (SBM), which has drift $b(x) = 0$ and diffusion $\sigma(x) = 1$. We first give a high-level description of the algorithm and our numerical experiments (Section 3.1), then we show how to formally derive the algorithm, using simple Taylor expansions (Section 3.2). In Section 3.3 we present our main convergence results, with proofs in Section 3.4. Section 3.5 proves a sufficient condition for regularity of the solution to the associated backward equation.

3.1. Overview of the simulation algorithm

Consider a SBM $\{Y_t\}_{t\in[0,T]}$, where T is the total length of time. We construct a numerical approximation $\{X_k\}_{k=1,\dots,N}$, where X_k is the value of the approximation at time $t_k = kh$, and h is our time step, so that $N = T/h$ is the total number of steps. We always assume that h is chosen such that N is an integer.

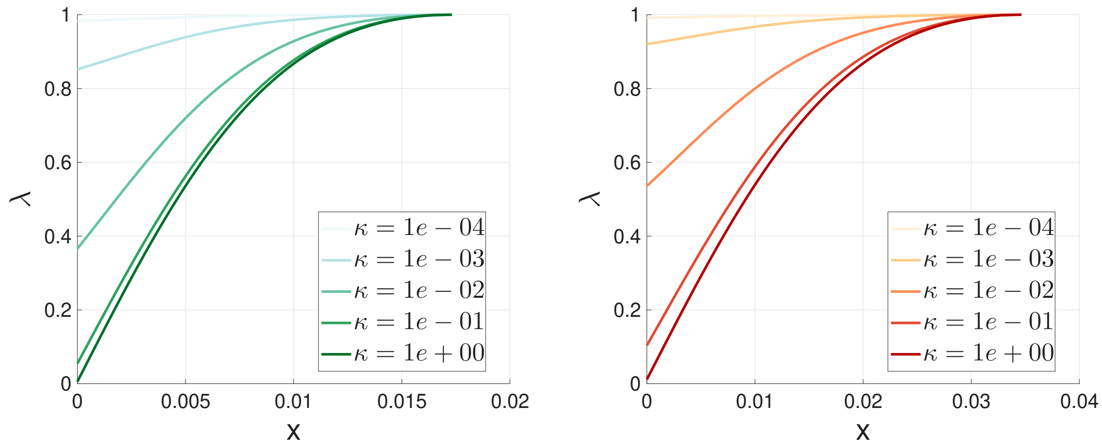

FIGURE 1. Plot of $\lambda(x)$ for $x \in [0, \sqrt{3h}]$ with various κ (shown in the legends) and time step $h = 10^{-4}$ (left) and $h = 4 \times 10^{-4}$ (right). We always have $\lambda(\sqrt{3h}) = 1$, while $\lambda(0)$ is a positive constant for all κ which decreases to 0 as κ increases. Increasing h has the effect of increasing $\lambda(\cdot)$.

The numerical scheme works as follows. We divide the domain into two regions:

$$\begin{aligned} \text{boundary layer:} \quad & \Omega_b = [0, \sqrt{3h}], \\ \text{exterior:} \quad & \Omega_e = (\sqrt{3h}, \infty). \end{aligned}$$

The algorithm proposes different updates, depending on whether X_k is in the boundary layer or the exterior.

Given a point in the *exterior*, $X_k \in \Omega_e$, it is updated with a standard Euler-Maruyama (EM) increment. Specifically,

$$X_{k+1} = X_k + Z_k, \tag{3.1}$$

where $\{Z_k\}_{k=0,\dots,N-1} \sim \mathcal{U}([-\sqrt{3h}, \sqrt{3h}])$ is a sequence of *iid* uniform random variables, which have mean 0 and variance $\mathrm{Var}(Z_k) = h$. We choose uniform random variables rather than normal random variables to simplify our later analysis. Notice that, because of the choice of uniform steps, if $X_k \in \Omega_e$, then $X_{k+1} > 0$, so we do not need to consider reflection at this stage.

Given a point X_k in the *boundary layer*, $X_k \in \Omega_b$, it does one of two things: with probability $\lambda(X_k)$ it is updated with a reflected EM increment [30], and with probability $1 - \lambda(X_k)$ it jumps exactly to 0:

$$X_{k+1} = \begin{cases} |X_k + Z_k|, & \text{with probability } \lambda(X_k) \\ 0, & \text{with probability } 1 - \lambda(X_k). \end{cases} \tag{3.2}$$

We remark that this same update is performed when $X_k = 0$; if the second case in (3.2) is chosen, the process simply stays at 0.

We will show momentarily that we achieve good convergence properties when λ is the following function:

$$\lambda(x) = \frac{x^2 + 2\kappa x + h}{\left(\frac{\kappa}{\sqrt{3h}} + 1\right) x^2 + h + \kappa\sqrt{3h}}. \tag{3.3}$$

Figure 1 shows $\lambda(x)$ for different values of κ and h. It has many sensible properties: one, λ is an increasing function of x, with $\lambda(\sqrt{3h}) = 1$; this is expected because when X_k is further away from 0, it should have a

Algorithm 1 Approximation of SBM X_T starting from $x \in [0, \infty)$.

Set $X_0 = x$.
for $k = 0$ **to** $N-1$ **do**
  Generate $Z_k \sim \mathcal{U}([-\sqrt{3h}, \sqrt{3h}])$.
  if $X_k > \sqrt{3h}$ **then**
    Update X_{k+1} using (3.1).
  else $(X_k \in [0, \sqrt{3h}])$
    Calculate $\lambda(X_k)$ using (3.3).
    Generate $u \sim \mathcal{U}([0,1])$.
    if $u \leq \lambda(X_k)$ **then**
      Update $X_{k+1} = |X_k + Z_k|$.
    else
      Set $X_{k+1} = 0$.
    end if
  end if
end for

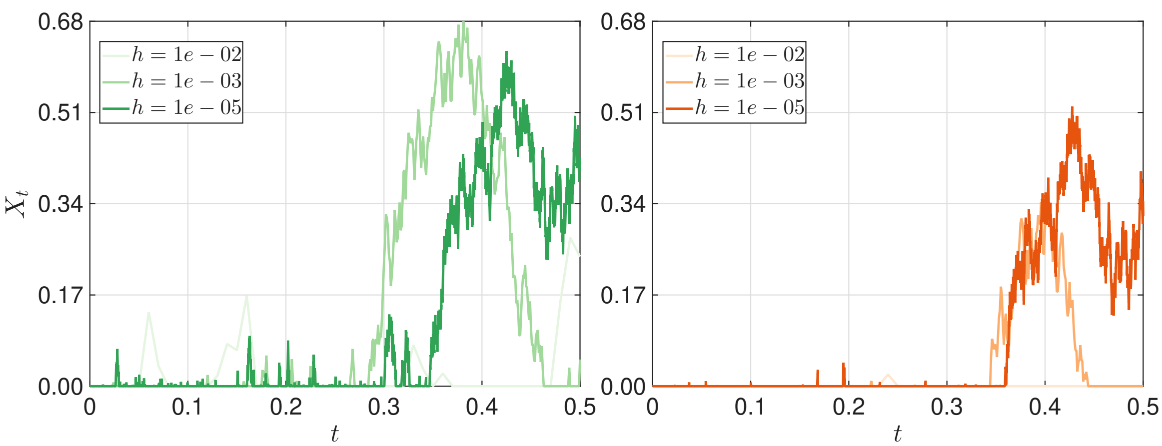

FIGURE 2. Simulations of an SBM with $X_0 = 0$ using Algorithm 1 with sticky parameter $\kappa = 0.6$ (left) and $\kappa = 3$ (right), and with time steps $h \in \{10^{-2}, 10^{-3}, 10^{-5}\}$ as shown in the legend. The same realizations for the random variables are used for both plots for the same h. Readers can observe how stickiness causes trajectories to spend more time near the sticky boundary, before beginning excursions.

smaller probability to jump there. Two, as κ increases, λ decreases, which also makes sense because when the boundary is stickier, there should be a greater chance of jumping there.

Our algorithm is summarized in algorithm 1. Some realizations from our algorithm are illustrated in Figure 2. We can see that as sticky parameter κ increases, realizations spend more time within the boundary layer.

Note that it is essential that λ be a valid probability, i.e. that $0 \leq \lambda(x) \leq 1$. This is not guaranteed in the derivation to follow, but is something that must be checked a posteriori.

Lemma 3.1. *The jump probability* λ *defined in* (3.3) *is a valid probability:* $\lambda(x) \in [0,1]$ *for all* $x \in [0, \sqrt{3h}]$.

Proof. By factoring part of the denominator, we can write λ as

$$\lambda(x) = \frac{x^2 + h + 2\kappa x}{x^2 + h + 2\kappa x + \frac{\kappa}{\sqrt{3h}}(x - \sqrt{3h})^2}.$$

From this is it clear that $\lambda(x) > 0$, since every term in the expression is positive. It is also clear that $\lambda(x) \leq 1$, with equality only when $x = \sqrt{3h}$ or when $\kappa = 0$. □

3.2. How to choose $\lambda(x)$

The key step of this algorithm is in choosing the function λ which governs the probability of jumping directly to 0, instead of performing an EM update. This is done by solving for a λ that makes the transition operator of the discrete scheme a good approximation to the transition operator of SBM over one timestep. In this section we derive a formula for λ formally, using Taylor expansions of the generator for SBM. Our intent is to show the key steps one must follow to apply these ideas more generally; we prove rigorous convergence statements in subsequent sections.

Let the transition operator of our numerical scheme with timestep h be denoted by P^h; it characterizes the evolution of statistics via

$$(P^h f)(x) = \mathbb{E}_x f(X_1).$$

We write $\mathbb{E}_x(\cdot) := \mathbb{E}(\cdot|X_0 = x)$. We wish to find λ such that P^h is a good approximation for the transition operator of the true dynamics; i.e. such that

$$(P^h f)(x) - f(x) \approx \mathbb{E}_x \int_0^h \mathcal{L}f(Y_s)ds\,, \tag{3.4}$$

where $\mathcal{L}f = \frac{1}{2}\partial_{xx} f$ for a SBM. We have that

$$\mathbb{E}_x \int_0^h \mathcal{L}f(Y_s)ds \approx h(\mathcal{L}f)(x),$$

and

$$\begin{aligned}(P^h f)(x) - f(x) &= \int_0^{+\infty} p_h(y|x)\left(f(t,y) - f(t,x)\right)\,dy \\ &\approx \partial_x f(t,x)\cdot\mathbb{E}_x(\Delta X) + \frac{1}{2}\partial_{xx} f(t,x)\cdot\mathbb{E}_x(\Delta X^2).\end{aligned}$$

Here $p_h(y|x)$ is the transition probability to move from x to y in one step of the scheme, and $\Delta X := X_1 - X_0$. Thus, asking that the approximations for both sides of (3.4) are within some tolerance say ϵ, our task is to find a λ such that

$$|\mathbb{E}_x(\Delta X)(\partial_x f) + \frac{1}{2}\mathbb{E}_x(\Delta X^2)(\partial_{xx} f) - \frac{h}{2}(\partial_{xx} f)| \leq \epsilon. \tag{3.5}$$

Later, we will show that choosing $\epsilon = O(h^2 + hx)$ gives good convergence properties.

We must choose λ such that the coefficients of $\partial_x f, \partial_{xx} f$ on the left-hand side of (3.5) are sufficiently small. This is not yet possible since there is only one term involving $\partial_x f$. Thus, our next step is to approximate $\partial_x f$ in terms of $\partial_{xx} f$, using the sticky boundary condition $\partial_x f(t,0) = \kappa\partial_{xx} f(t,0)$. Taylor-expanding $\partial_x f, \partial_{xx} f$ around x shows that

$$\begin{aligned}\partial_x f(t,0) &= \partial_x f(t,x) - x\partial_{xx} f(t,x) + \frac{x^2}{2} f^{(3)}(t,z_x), \\ \partial_{xx} f(t,0) &= \partial_{xx} f(t,x) - x f^{(3)}(t,y_x),\end{aligned}$$

for some $z_x, y_x \in [0,x]$. From the sticky boundary condition, we have

$$\partial_x f(t,x) = \kappa \partial_{xx} f(t,x) + x\partial_{xx} f(t,x) - \kappa x f^{(3)}(t,y_x) - \frac{x^2}{2} f^{(3)}(t,z_x). \tag{3.6}$$

Therefore, an approximation of $\partial_x f(t,x)$ using only $\partial_{xx} f(t,x)$ is

$$\partial_x f(t,x) = (\kappa + x)\partial_{xx} f(t,x) + O(h+x). \tag{3.7}$$

We remark that to obtain an error of order $O(h+x)$, we do not need the additional term $x\partial_{xx} f(t,x)$; in fact, this term has the same magnitude as some higher-order terms that we are dropping. However, we found that without including this term, we obtained a function λ that was not always a valid probability, i.e. $\lambda \notin [0,1]$.

Substituting for $\partial_x f$ into (3.5), shows that we must find a function λ such that

$$|(\kappa + x)\mathbb{E}_x(\Delta X) + \frac{1}{2}\mathbb{E}_x(\Delta X^2) - \frac{h}{2}| \approx 0. \tag{3.8}$$

At this point we may evaluate $\mathbb{E}_x(\Delta X), \mathbb{E}_x(\Delta X^2)$, and then solve (3.8) for λ, treating (3.8) as an equality. We have that

$$p_h(y|x) = (1-\lambda)1_{\{0\}}(y) + \frac{\lambda}{2\sqrt{3h}} 1_{[0,x+\sqrt{3h}]}(y) + \frac{\lambda}{2\sqrt{3h}} 1_{[0,|x-\sqrt{3h}|]}(y).$$

From this, we can calculate

$$\begin{aligned} \mathbb{E}_x(\Delta X_k) &= -x + \frac{\lambda\sqrt{3h}}{2} + \frac{\lambda x^2}{2\sqrt{3h}}\,, \\ \mathbb{E}_x(\Delta X_k^2) &= -\frac{\lambda x^3}{\sqrt{3h}} + (1+\lambda)x^2 - \sqrt{3h}\lambda x + \lambda h\,. \end{aligned} \tag{3.9}$$

Substituting into (3.8) and solving for λ gives gives (3.3).

Our calculations above have demonstrated two lemmas, which we will use shortly in our convergence proofs.

Lemma 3.2. *Consider a function $u(t,x)$ satisfying the sticky boundary condition at $x = 0$, i.e. such that $\frac{1}{2}\partial_x u(t,0) = \kappa\partial_{xx} u(t,0)$. Given $x \in \Omega_b$ and $t \in [0,+\infty)$, there exists $z_x, y_x \in [0,x]$ such that*

$$\partial_x u(t,x) = (\kappa + x)\cdot\partial_{xx} u(t,x) - \kappa x u^{(3)}(t,y_x) - \frac{x^2}{2} u^{(3)}(t,z_x). \tag{3.10}$$

Proof. This follows from our calculations deriving (3.6). ☐

Lemma 3.3. *For λ chosen as (3.3), and given $x \in \Omega_b$,*

$$(\kappa + x)\mathbb{E}_x(\Delta X) + \frac{1}{2}\mathbb{E}_x(\Delta X^2) - \frac{h}{2} = 0. \tag{3.11}$$

Proof. This follows from our calculations following (3.8), which showed that we chose λ such that the above equation holds. ☐

3.3. Primary convergence results

Now we give an overview of our main convergence results. We wish to show weak convergence of the numerical scheme with order h. That is, we wish to show that, for h sufficiently small, there is a constant $C > 0$ such that

$$|\mathbb{E}\phi(X_N) - \mathbb{E}\phi(Y_T)| \le Ch, \tag{3.12}$$

where ϕ is from a suitable class of functions. This is shown by decomposing the error into a local, one-step error, which is different for the boundary and the interior, and a contribution related to the number of steps accumulating each type of error. Our proof strategy closely follows that used to show convergence of diffusions satisfying Robin boundary conditions [30].

Throughout the convergence analysis, we take $X_0 = Y_0 = x_0$ deterministically and suppress the dependence of expectations on x_0 unless it must be emphasized. The SBM bounds below are uniform in x_0, whereas the general sticky-diffusion bound in Section 4 contains the factor $1+x_0^4$. A random initial condition can be handled by conditioning on X_0, provided the resulting x_0-dependent bound is integrable.

As is typical in proofs of weak convergence [30, 37], our main tool will be the backward equation (2.4), repeated here for an SBM for ease of reference:

$$\begin{gathered}\partial_t u + \frac{1}{2}\partial_{xx}^2 u = 0 \qquad (x \geq 0), \\ \partial_x u(t,0) = \kappa \cdot \partial_{xx}^2 u(t,0), \quad u(T,x) = \phi(x).\end{gathered} \tag{3.13}$$

We may observe that for $X_0 = x_0$,

$$\mathbb{E}(\phi(Y_T)) = u(0,x_0), \qquad \mathbb{E}(\phi(X_N)) = \mathbb{E}(u(T,X_N)),$$

where the latter comes from observing that $u(T,X_N) = \mathbb{E}[\phi(Y_T)|Y_T = X_N] = \phi(X_N)$. Therefore, writing $t_k = kh$, $u_k = u(t_k, X_k)$, we have

$$\mathbb{E}\left(\phi(X_N)\right) - \mathbb{E}\left(\phi(Y_T)\right) = \mathbb{E}\left(\sum_{k=0}^{N-1} u_{k+1} - u_k\right). \tag{3.14}$$

Thus, the global error can be decomposed into a sum of local errors, via:

$$\begin{aligned}|\mathbb{E}\phi(X_N) - \mathbb{E}\phi(Y_T)| &= \left|\mathbb{E}\left(\sum_{k=0}^{N-1}\mathbb{E}\left(u_{k+1} - u_k|X_k\right)\right)\right| \\ &\leq \mathbb{E}\left(\sum_{k=0}^{N-1}\left|\mathbb{E}\left(u_{k+1} - u_k|X_k\right)\right| \cdot \mathbf{1}_{X_k\in[0,\sqrt{3h}]}\right) + \mathbb{E}\left(\sum_{k=0}^{N-1}\left|\mathbb{E}\left(u_{k+1} - u_k|X_k\right)\right| \cdot \mathbf{1}_{X_k\notin[0,\sqrt{3h}]}\right).\end{aligned} \tag{3.15}$$

The global error thus arises from two things: the local errors $|\mathbb{E}\left(u_{k+1} - u_k \mid X_k\right)|$ in the boundary and exterior, and the number of steps in each of these states.

To present our main results we need some assumptions.

Assumption 1. The solution $u(t,x)$ of (3.13) has uniformly bounded spatial derivatives up to order 4. That is,

$$\|\partial_x^j u\|_\infty := \sup_{(t,x)\in[0,T]\times[0,\infty)} \left|\partial_x^j u(t,x)\right| < \infty, \qquad j = 1,2,3,4.$$

Assumption 2. The solution $u(t,x)$ of (3.13) has time derivative up to order 2 where $\|\partial_{tt}u\|_\infty < \infty$.

We further suppose that h is always chosen such that $N := T/h \in \mathbb{Z}$.

Theorem 3.4. *(Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in Algorithm 1)*
Under Assumptions 1, 2, there exists a constant M depending on $\left\|u^{(3)}\right\|_\infty$ $\left\|u^{(4)}\right\|_\infty$ and $\|\partial_{tt}u\|_\infty$, such that

$$|\mathbb{E}\left(u_{k+1} - u_k|X_k = x\right)| \leq \begin{cases} M(h^2 + hx), & \forall x \in [0,\sqrt{3h}] \\ Mh^2 & \forall x > \sqrt{3h}. \end{cases} \tag{3.16}$$

Thus, the local error of the scheme in the boundary layer is $O(h^{3/2})$ and in the exterior is $O(h^2)$.

A naive estimate for the global error would be $O(h^{1/2})$, which comes from taking the worst-case local error of $O(h^{3/2})$, and multiplying by the total number of steps which is $O(h^{-1})$. This however is too pessimistic, as will turn out that the number of steps in the boundary layer is $O(h^{-1/2})$, leading to a global error of $O(h^{3/2} \cdot h^{-1/2}) + O(h^2 \cdot h^{-1}) \sim O(h)$.

The specific type of estimate we must construct on the number of steps in the boundary layer arises from this corollary.

Corollary 3.4.1. *Under the assumptions of Theorem 3.4,*

$$\mathbb{E}\left(\sum_{k=0}^{N-1} |\mathbb{E}\left(u_{k+1}-u_k|X_k\right)| \cdot \mathbf{1}_{X_k\in[0,\sqrt{3h}]}\right) \le M_1\left(h + h\cdot\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \cdot \mathbf{1}_{X_k\in[0,\sqrt{3h}]}\right)\right).$$

Proof. This is simply the expectation applied to the first case of (3.16), and summed over all steps. □

Our main lemma to estimate the number of points in the boundary layer is the following.

Lemma 3.5. *For sufficiently small h, there exists a constant $C > 0$, independent of h and initial position $x_0 \ge 0$, such that*

$$\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \cdot \mathbf{1}_{X_k\in[0,\sqrt{3h}]}\right) < C.$$

This lemma implies the number of points in the boundary layer is $O(h^{-1/2})$, because $|X_k| = O(h^{1/2})$ in the boundary layer. It is the most technical lemma to prove.

Applying this Lemma to (3.15) gives the global error bound, our main convergence theorem.

Theorem 3.6. *(Global Error of $\{X_k\}_{k\in\mathbb{N}}$ in Algorithm 1)*
Let $\phi : [0,\infty) \to \mathbb{R}$ be such that the associated backward solution of (3.13),

$$u(t,x) := \mathbb{E}[\phi(Y_T) \mid Y_t = x]\,,$$

satisfies Assumptions 1 and 2. Then there exist constants $\delta > 0$ and $C > 0$, independent of h and initial position $x_0 \ge 0$, such that for all $h < \delta$ and $x_0 \ge 0$,

$$\left|\mathbb{E}\big(\phi(X_N) - \phi(Y_T)\big)\right| \le Ch. \tag{3.17}$$

Remark. The main convergence theorems are stated in terms of regularity assumptions on the backward solution u, rather than directly on the terminal datum ϕ. A sufficient condition on ϕ ensuring Assumptions 1 and 2 is given later in Section 3.5; see in particular Proposition 3.10. For simplicity, that sufficient condition is stated on a compact interval $[0, L]$ with a reflecting boundary condition at $x = L$ (general Wentzell conditions). In fact, at 0 and L, we can generalize the proof of Proposition 3.10 to any Wentzell condition.

Proof. This follows from (3.15), applying Theorem 3.4, Corollary 3.4.1, and Lemma 3.5. □

Thus, our scheme is weakly convergent with order 1.

Numerical experiments. The convergence results are illustrated numerically in Figure 3. We run our algorithm with $X_0 = 0, T = 1$ and estimate $\mathbb{E}[X_T]$. In our algorithm we additionally impose a reflecting boundary condition at $x = L$, which is implemented in our algorithm using a reflected EM update (similar to the update in our algorithm at $x = 0$). This is known to converge weakly with order 1 [30]. We compare our estimated mean, to the solution $u(0,0)$ to the backward equation (3.13) with $\phi(x) = x$. This solution is computed using a finite-difference scheme with with extremely small space and time steps; specifically, $\Delta x = 10^{-3}$ and $\Delta t = \frac{\Delta x^2}{8*5} = 2.5 \times 10^{-8}$.

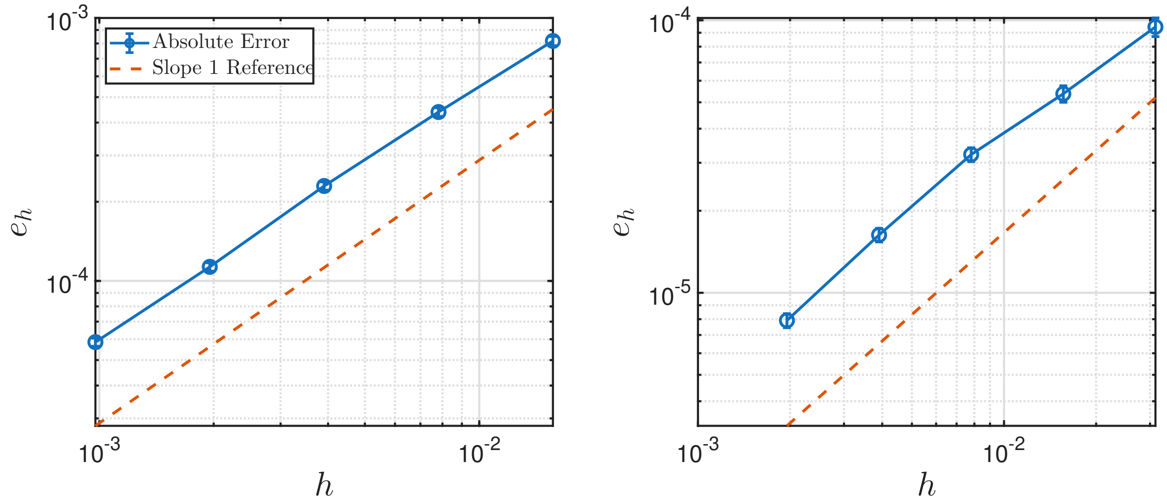

Figure 3. Log-log plot for the weak error $e_h = |\mathbb{E}(X_T|X_0 = 0) - u(0,0)|$ of our simulation scheme 1 using $\kappa = 1$ (left) and $\kappa = 10$ (right). Both comparisons used $L = 2, T = 1$. Error bars are ± 1 empirical standard deviation. The left plot takes time step-size $h \in \{1/64, 1/128, 1/256, 1/512, 1/1024\}$. To keep the relative error bar similar, we quadruple the number of realizations every time we halve the time step size starting from 2×10^8 realization. The least square fit gives the slope of 0.956615. The right plot takes time step-size $h \in \{1/32, 1/64, 1/128, 1/256, 1/512\}$. Similarly, to keep the relative error bar similar, we start with 8×10^8 realizations and we quadruple the number of realizations every time we halve the time step size. The least square fit for the last three data points has the slope of 1.011378.

Figure 3 demonstrates $O(h)$ convergence for two different values of the sticky parameter. It is notable that our choice of ϕ does not satisfy the compatibility conditions given in Proposition 3.10, which are used to show that Assumptions 1 and 2 hold for the solution to the backward equation. Nevertheless, this does not appear to affect our empirical order of convergence.

3.4. **Convergence proofs**

It remains to prove Theorem 3.4 and Lemma 3.5. We start with a couple of lemmas regarding the magnitudes of the moments of ΔX_k. In this section, we use notation $\mathbb{E}_x f(\Delta X_k) \equiv \mathbb{E}[f(\Delta X_k)|X_k = x]$ for a function f of the increments.

Lemma 3.7. *Given the sequence $\{X_k\}$ generated from algorithm 1 with $\kappa > 0$, there is a constant C such that*

$$\sup_{x\in[0,\infty)} |\mathbb{E}_x(\Delta X_k^2)| \leq Ch\,, \qquad \sup_{x\in[0,\sqrt{3h}]} |\mathbb{E}_x(\Delta X_k)| \leq Ch.$$

The second part of the lemma is not trivial, as we might expect $|\mathbb{E}_x(\Delta X_k)| \sim O(h^{1/2})$, since this is the size of a typical jump to 0.

Proof. The bound on $\mathbb{E}_x(\Delta X_k^2)$ follows trivially from the fact that $|\Delta X_k^2| \leq 3h$. The bound on $\mathbb{E}_x(\Delta X_k)$ then follows from Lemma 3.3, which shows that $\mathbb{E}_x(\Delta X_k)$ is a sum of terms which are all $O(h)$ or higher provided $\kappa > 0$. □

Lemma 3.8. *There exists a constant C such that, for every $x \in [0, \sqrt{3h}]$,*

$$\mathbb{E}_x(|\Delta X_k|^3) \leq C(h^2 + hx).$$

Proof. Fix $x \in [0, \sqrt{3h}]$. Positive increments occur only on the continuous branch and satisfy $\Delta X_k^+ \leq \sqrt{3h}$, while $X_{k+1} \geq 0$ implies $\Delta X_k^- \leq x$. Hence

$$\mathbb{E}_x|\Delta X_k|^3 \leq \lambda(x)(3h)^{3/2} + x^3.$$

For $x \leq \sqrt{3h}$, the numerator and denominator in (3.3) satisfy

$$x^2 + 2\kappa x + h \leq 4h + 2\kappa x, \qquad \left(\frac{\kappa}{\sqrt{3h}} + 1\right) x^2 + h + \kappa\sqrt{3h} \geq \kappa\sqrt{3h}.$$

Therefore

$$\lambda(x) \leq \frac{4\sqrt{h}}{\sqrt{3}\,\kappa} + \frac{2x}{\sqrt{3h}},$$

and consequently

$$\mathbb{E}_x|\Delta X_k|^3 \leq C(h^2 + hx) + x^3 \leq C'(h^2 + hx),$$

where the last inequality follows from $x^2 \leq 3h$, and hence $x^3 \leq 3hx$. □

We are now ready to prove the local error.

Proof of Theorem 3.4. Let u be the solution to the backward equation (3.13). We perform a Taylor expansion of u to 3rd order centered at (t_k, X_k). We write $u_k = u(t_k, X_k), \partial_x u_k = \partial_x u(t_k, X_k)$ (and similarly for $\partial_{xx} u_k, \partial_{xxx} u_k$), and we write Y_k for a point such that $Y_k \in [X_k, X_{k+1}]$ or $Y_k \in [X_{k+1}, X_k]$, and s_k will be a point in interval $s_k \in [kh, (k+1)h]$). Expanding gives

$$\begin{aligned} u_{k+1} - u_k = \partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2}\partial_{xx} u_k \cdot \Delta X_k^2 \\ + \frac{1}{2}\partial_{tt} u(s_k, X_k) \cdot h^2 + \frac{1}{6}\partial_{xxx} u(t_k, Y_k) \cdot \Delta X_k^3. \end{aligned} \tag{3.18}$$

We are interested in $\mathbb{E}_x(u_{k+1} - u_k)$. We start by assuming $x \in \Omega_b$. We now bound the expectation of each of the terms in the Taylor-expansion (3.18). We have

$$\begin{aligned} \mathbb{E}_x|\partial_{tt} u(s_k, X_k) \cdot h^2| \leq h^2 \|\partial_{tt} u\|_\infty \\ \mathbb{E}_x|\partial_{xxx} u(t_k, Y_k) \cdot \Delta X_k^3| \leq C(h^2 + hx)\|u^{(3)}\|_\infty, \end{aligned}$$

where the second inequality follows from Lemma 3.8.

We also have, using the fact that u solves the backward equation (3.13) and using Lemma 3.2,

$$\begin{aligned} &\mathbb{E}_x\left[\partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2}\partial_{xx} u_k \cdot (\Delta X_k^2)\right] \\ &= -\frac{1}{2}\partial_{xx} u(t_k, x) \cdot h + (\kappa + x) \cdot \partial_{xx} u(t_k, x) \cdot \mathbb{E}_x(\Delta X_k) + \frac{1}{2}\partial_{xx} u(t_k, x) \cdot \mathbb{E}_x(\Delta X_k^2) \\ &\quad - \mathbb{E}_x(\Delta X_k)\left(\kappa x \cdot u^{(3)}(t_k, y_x) + \frac{x^2}{2} \cdot u^{(3)}(t_k, z_x)\right) \\ &= -\mathbb{E}_x(\Delta X_k)\left(\kappa x \cdot u^{(3)}(t_k, y_x) + \frac{x^2}{2} \cdot u^{(3)}(t_k, z_x)\right). \end{aligned} \tag{3.19}$$

Here $y_x, z_x \in [0, x]$, and the last step used Lemma 3.3.

Therefore, applying the triangle inequality to (3.18), and then using Lemma 3.7, shows that there exist constants M_1, M_2, independent of x, t, h, such that

$$\begin{aligned}|\mathbb{E}\left(u_{k+1}-u_k|X_k=x\right)| &\leq |\mathbb{E}_x(\Delta X_k)|\cdot\left(\kappa x\|u^{(3)}\|_\infty+\frac{h}{2}\|u^{(3)}\|_\infty\right)+h^2\|\partial_{tt}u\|_\infty+C(h^2+hx)\|u^{(3)}\|_\infty\\ &\leq M_1h^2+M_2hx.\end{aligned}$$

We now check $x\in\Omega_e$. Recall that, outside the boundary layer, the simulation scheme makes a standard Euler Maruyama jump, with moments

$$\mathbb{E}_x(\Delta X_k)=\mathbb{E}_x(\Delta X_k^3)=0\,,\quad \mathbb{E}_x(\Delta X_k^2)=h,\quad \mathbb{E}_x(\Delta X_k^4)=\frac{9}{5}h^2.$$

We exploit the Taylor expansion up to fourth order,

$$\begin{aligned}u_{k+1}-u_k &= \partial_t u_k\cdot h+\partial_x u_k\cdot\Delta X_k+\frac{1}{2}\partial_{xx}u_k\cdot\Delta X_k^2\\ &\quad+\frac{1}{2}\partial_{tt}u(s_k,X_k)\cdot h^2+\frac{1}{6}\partial_{xxx}u_k\cdot\Delta X_k^3+\frac{1}{24}\partial_{xxxx}u(t_k,Y_k)\cdot\Delta X_k^4.\end{aligned}\tag{3.20}$$

Using the fact that u solves the backward equation (3.13), we know,

$$\begin{aligned}\mathbb{E}_x\left[\partial_t u_k\cdot h+\partial_x u_k\cdot\Delta X_k+\frac{1}{2}\partial_{xx}u_k\cdot(\Delta X_k^2)\right] &= -\frac{1}{2}\partial_{xx}u(t_k,x)\cdot h+\partial_x u_k\cdot 0+\frac{1}{2}\partial_{xx}u(t_k,x)\cdot h\\ &=0.\end{aligned}$$

Apply the Taylor expansion (3.18) and the fact that $\mathbb{E}_x(\Delta X_k^3)=0$,

$$\begin{aligned}|\mathbb{E}\left(u_{k+1}-u_k|X_k=x\right)| &= \left|\frac{1}{2}\partial_{tt}u(s_k,x)\cdot h^2+0+\frac{3}{40}\partial_{xxxx}u(t_k,Y_k)\cdot h^2\right|\\ &\leq h^2\left(\|\partial_{tt}u\|_\infty+\|\partial_{xxxx}u\|_\infty\right).\end{aligned}$$

□

It remains to prove Lemma 3.5. For this, we first record a standard discrete backward equation for additive functionals of a Markov chain, proven here for completeness [30, 37].

Lemma 3.9. *Let $\{X_i\}_{i=0}^N$ be the Markov chain generated by our numerical scheme, with time grid $t_i = ih$, $i = 0,\dots,N$, and $t_N = T$. Let P denote the transition operator of the scheme, i.e. whenever the conditional expectation is well defined,*

$$PU(t_i,x) := \mathbb{E}\big[U(t_{i+1},X_{i+1})\,\big|\,X_i=x\big],\qquad i=0,\dots,N-1.$$

Let $g:\{t_0,\dots,t_{N-1}\}\times[0,\infty)\to\mathbb{R}$ be measurable, and assume that for every $i=0,\dots,N$ and every $x\in[0,\infty)$,

$$\mathbb{E}\left(\sum_{k=i}^{N-1}|g(t_k,X_k)|\,\middle|\,X_i=x\right)<\infty.$$

Define

$$U(t_i,x) := \mathbb{E}\left(\sum_{k=i}^{N-1}g(t_k,X_k)\,\middle|\,X_i=x\right),\qquad i=0,\dots,N,$$

with the convention that the sum is empty when $i = N$. Then U satisfies

$$\begin{aligned} PU(t_i, x) - U(t_i, x) &= -g(t_i, x), \qquad (t_i, x) \in \{t_0, \dots, t_{N-1}\} \times [0, \infty), \\ U(t_N, x) &= 0, \qquad x \in [0, \infty). \end{aligned} \tag{3.21}$$

Proof. By assumption, the random variable $\sum_{k=i}^{N-1} g(t_k, X_k)$ is integrable under $\mathbb{P}(\cdot \mid X_i = x)$, so $U(t_i, x)$ is well defined. When $i = N$, the sum is empty, hence

$$U(t_N, x) = 0, \qquad x \in [0, \infty).$$

Now fix $i \in \{0, \dots, N-1\}$. Since $X_i = x$ on the conditioning event,

$$U(t_i, x) = g(t_i, x) + \mathbb{E}\left(\sum_{k=i+1}^{N-1} g(t_k, X_k) \,\middle|\, X_i = x \right).$$

Applying the tower property,

$$\begin{aligned} \mathbb{E}\left(\sum_{k=i+1}^{N-1} g(t_k, X_k) \,\middle|\, X_i = x \right) &= \mathbb{E}\left(\mathbb{E}\left(\sum_{k=i+1}^{N-1} g(t_k, X_k) \,\middle|\, X_{i+1} \right) \,\middle|\, X_i = x \right) \\ &= \mathbb{E}\big[U(t_{i+1}, X_{i+1}) \,\big|\, X_i = x\big] \\ &= PU(t_i, x). \end{aligned}$$

Therefore,

$$U(t_i, x) = g(t_i, x) + PU(t_i, x).$$

□

Now, to prove Lemma 3.5, it suffices to show that

$$U(t_0, x) = \mathbb{E}\left(\sum_{k=0}^{N-1} X_k \mathbf{1}_{\{X_k \in [0, \sqrt{3h}]\}} \,\middle|\, X_0 = x \right)$$

is uniformly bounded. From Lemma 3.9, we know that U solves (3.21) with $g(t, x) = x \cdot \mathbf{1}_{x \in [0, \sqrt{3h}]}$. Therefore, it is enough to construct a uniformly bounded function V such that

$$\begin{aligned} PV(t_i, x) - V(t_i, x) &\le -x \mathbf{1}_{\{x \in [0, \sqrt{3h}]\}}, \qquad i = 0, \dots, N-1, \\ V(T, x) &\ge 0. \end{aligned}$$

Indeed, these two inequalities imply $U(t_i, x) \le V(t_i, x)$ for all grid times by backward induction (shown explicitly in the proof), and in particular

$$U(t_0, x) \le V(t_0, x).$$

Hence the uniform boundedness of V yields the desired bound on U.

Proof of Lemma 3.5. Following the discrete supersolution construction in [30], specialized to $q \equiv 1$, we construct a bounded spatial barrier whose one-step drift controls the boundary occupation term. A time-dependent exponential factor will absorb the order-h remainder in this drift estimate.

Fix $M, s > 16$ and define

$$w(x) = \begin{cases} M, & x \in [0, \sqrt{3h}/2], \\ M - s\left(x - \dfrac{\sqrt{3h}}{2} \right), & x \in (\sqrt{3h}/2, \infty). \end{cases} \tag{3.22}$$

The case-by-case calculation in Supplement A.1 gives constants $C_1 > 0$ and $\delta > 0$, independent of x and h, such that, for every $h < \delta$,

$$Pe^{w(x)} \le e^{w(x)}\big(1 - f_h(x) + C_1 h\big), \qquad x \ge 0, \tag{3.23}$$

where f_h is defined by (A.1)–(A.5). Those formulas, together with $0 \le \lambda \le 1$, also give

$$x\mathbf{1}_{\{x\in[0,\sqrt{3h}]\}} \le f_h(x) \le C_2\sqrt{h}, \qquad x \ge 0, \tag{3.24}$$

for some $C_2 > 0$.

Define

$$V(t_i, x) := e^{K(T-t_i)}e^{w(x)}, \qquad i = 0, \dots, N. \tag{3.25}$$

Constants below may depend on the fixed parameters M, s and on K, but not on x or h. Using (3.23) and $e^{-Kh} = 1 - Kh + O_K(h^2)$, we obtain

$$\begin{aligned} PV(t_i,x) - V(t_i,x) &\le V(t_i,x)\left[-f_h(x) + (C_1 - K)h + C_K\big(hf_h(x) + h^2\big)\right] \\ &\le -V(t_i,x)\left[f_h(x) + (K - C_1)h - C_K' h^{3/2}\right], \end{aligned}$$

where the second inequality uses (3.24) and $h \le 1$.

Choose $K = C_1 + 1$ and decrease δ, if necessary, so that

$$(K - C_1)h - C_K' h^{3/2} \ge 0, \qquad h < \delta.$$

Then

$$PV(t_i,x) - V(t_i,x) \le -V(t_i,x)f_h(x).$$

After decreasing δ once more,

$$w(x) \ge M - \frac{s}{2}\sqrt{3h} \ge 0, \qquad x \in [0, \sqrt{3h}],$$

and hence $V(t_i, x) \ge 1$ on the boundary layer. Together with (3.24), this gives

$$PV(t_i,x) - V(t_i,x) \le -x\mathbf{1}_{\{x\in[0,\sqrt{3h}]\}}. \tag{3.26}$$

For the function U defined immediately before the proof, Lemma 3.9 gives

$$PU(t_i,x) - U(t_i,x) = -x\mathbf{1}_{\{x\in[0,\sqrt{3h}]\}}, \qquad U(T,x) = 0.$$

Moreover, $V(T,x) = e^{w(x)} \ge 0$. We therefore prove $U(t_i,x) \le V(t_i,x)$ by backward induction. Indeed, if the inequality holds at time t_{i+1}, monotonicity of P gives $PU(t_i,x) \le PV(t_i,x)$, and hence

$$\begin{aligned} U(t_i,x) &= PU(t_i,x) + x\mathbf{1}_{\{x\in[0,\sqrt{3h}]\}} \\ &\le PV(t_i,x) + x\mathbf{1}_{\{x\in[0,\sqrt{3h}]\}} \\ &\le V(t_i,x), \end{aligned}$$

where the last inequality follows from (3.26).

Finally, $w(x) \le M$ for every $x \ge 0$, so

$$U(t_0,x_0) \le V(t_0,x_0) \le e^{KT+M}, \qquad x_0 \ge 0.$$

Thus the conclusion holds with $C = e^{KT+M}$, which is independent of h and x_0. □

As a side remark, we in fact use a special case of 3.19-3.21 in Leimkuhler's paper [30] where $q(\cdot) = 1$.

3.5. **Sufficient conditions for backward equation regularity**

The convergence theorems assume regularity of the backward solution u. We record a concrete sufficient condition on the terminal datum ϕ on a compact interval $[0, L]$, where standard regularity theory for the sticky/Wentzell problem is available. We impose reflection at $x = L$ for simplicity; Remark 3.5 gives the corresponding general Wentzell condition.

The compatibility conditions below place ϕ in the domain $D(A^2)$ of the squared spatial generator: the first pair gives $\phi \in D(A)$, and the second gives $A\phi \in D(A)$. Analytic-semigroup regularity then yields the required uniform bounds.

Proposition 3.10. *Let $\phi \in C^4([0, L])$, and let $u(t, x)$ solve*

$$\partial_t u + \frac{1}{2}\partial_{xx} u = 0, \qquad (t, x) \in [0, T) \times [0, L], \tag{3.27a}$$
$$\partial_x u(t, 0) = \kappa\, \partial_{xx} u(t, 0), \qquad t \in [0, T), \tag{3.27b}$$
$$\partial_x u(t, L) = 0, \qquad t \in [0, T), \tag{3.27c}$$
$$u(T, x) = \phi(x), \qquad x \in [0, L]. \tag{3.27d}$$

Assume that ϕ satisfies compatibility conditions

$$\phi'(0) = \kappa\phi''(0), \qquad \phi'(L) = 0,$$

and

$$\phi^{(3)}(0) = \kappa\phi^{(4)}(0), \qquad \phi^{(3)}(L) = 0.$$

Then

$$\sup_{(t,x)\in[0,T]\times[0,L]} |\partial_x^j u(t, x)| < \infty, \qquad j = 1, 2, 3, 4,$$

and

$$\sup_{(t,x)\in[0,T]\times[0,L]} |\partial_{tt} u(t, x)| < \infty.$$

In particular, Assumptions 1 and 2 hold on the compact strip $[0, T] \times [0, L]$.

Proof. Set $v(s, x) := u(T - s, x)$. Then $v_s = \frac{1}{2}v_{xx}$ with the sticky condition at 0, reflection at L, and $v(0) = \phi$. On $X := C([0, L])$, define

$$Af := \frac{1}{2}f'', \qquad D(A) := \left\{f \in C^2([0, L]) : f'(0) = \kappa f''(0),\ f'(L) = 0\right\}.$$

By Theorem 1.1 of [14], A generates an analytic semigroup $\{T(s)\}_{s\geq 0}$ on X, so $v(s) = T(s)\phi$.

The first pair of compatibility conditions gives $\phi \in D(A)$; since $A\phi = \frac{1}{2}\phi''$, the second pair gives $A\phi \in D(A)$. Thus $\phi \in D(A^2)$. The standard differentiability theorem for C_0-semigroups [40, Section 2.4] gives

$$v_s(s) = T(s)A\phi, \qquad v_{ss}(s) = T(s)A^2\phi, \qquad Av(s) = T(s)A\phi, \qquad A^2 v(s) = T(s)A^2\phi.$$

Strong continuity implies that $\|T(s)f\|_\infty \leq M_T\|f\|_\infty$ for $0 \leq s \leq T$. Since $Av = \frac{1}{2}v_{xx}$ and $A^2 v = \frac{1}{4}v_{xxxx}$ in the interior,

$$\|v_{ss}(s, \cdot)\|_\infty \leq M_T\|A^2\phi\|_\infty,$$
$$\|v_{xx}(s, \cdot)\|_\infty \leq 2M_T\|A\phi\|_\infty,$$
$$\|v_{xxxx}(s, \cdot)\|_\infty \leq 4M_T\|A^2\phi\|_\infty.$$

Moreover, $v(s) \in D(A)$ and $Av(s) \in D(A)$ imply

$$v_x(s,0) = \kappa v_{xx}(s,0), \qquad v_{xxx}(s,0) = \kappa v_{xxxx}(s,0).$$

Integration from 0 to x therefore gives

$$|v_x(s,x)| \le (\kappa + L)\|v_{xx}(s,\cdot)\|_\infty,$$
$$|v_{xxx}(s,x)| \le (\kappa + L)\|v_{xxxx}(s,\cdot)\|_\infty.$$

Hence v has uniformly bounded spatial derivatives through order four and a uniformly bounded second time derivative on $[0,T] \times [0,L]$. Since $u(t,x) = v(T-t,x)$, the same bounds hold for u. $\square$

Remark. The same argument applies if the reflecting condition at $x = L$ is replaced by

$$\frac{1}{2}\partial_{xx}u(t,L) + \beta_L \partial_x u(t,L) + \gamma_L u(t,L) = 0.$$

In that case,

$$D(A) := \left\{ f \in C^2([0,L]) : f'(0) = \kappa f''(0),\ \frac{1}{2}f''(L) + \beta_L f'(L) + \gamma_L f(L) = 0 \right\},$$

and the compatibility conditions at L become

$$\frac{1}{2}\phi''(L) + \beta_L \phi'(L) + \gamma_L \phi(L) = 0,$$
$$\frac{1}{4}\phi^{(4)}(L) + \frac{\beta_L}{2}\phi^{(3)}(L) + \frac{\gamma_L}{2}\phi''(L) = 0.$$

4. General Sticky Diffusion

Now we consider a general one-dimensional sticky diffusion with drift $b(x)$ and diffusion $\sigma(x)$. For ease of reference, its generator is

$$\mathcal{L}u = b \cdot \partial_x u + a\partial_{xx}u, \tag{4.1}$$

with $a = \frac{\sigma^2}{2}$, and with a sticky boundary condition at $x = 0$:

$$a\partial_x u|_{x=0} = \kappa\left(b\partial_x u + a\partial_{xx}u\right)|_{x=0}. \tag{4.2}$$

The backward equation describing the process is

$$\partial_t u + \mathcal{L}u = 0, \quad x \in (0,+\infty), \qquad u(T,x) = \phi(x), \quad x \in (0,+\infty), \tag{4.3}$$

with the sticky boundary condition (4.2).

4.1. Overview of the simulation algorithm

The construction follows the SBM scheme, with three modifications. The variable coefficient $\sigma(x)$ makes the boundary layer implicit and potentially disconnected, although Lemma 4.1 localizes it to an $O(\sqrt{h})$ neighborhood of the origin for sufficiently small h. A nested reflection is used so that the jump probability is valid, and negative drift can create an additional thin reflection region. These features are addressed in the validity and convergence analysis below.

We generate $\{X_k\}_{k=0}^N$ with time step h. Given $X_k = x$, define

$$\begin{aligned} \text{boundary layer:} \quad & \Omega_b = \{x : 0 \le \frac{x}{\sigma(x)} \le \sqrt{3h}\}, \\ \text{exterior:} \quad & \Omega_e = \{x : \frac{x}{\sigma(x)} > \sqrt{3h}\}. \end{aligned} \tag{4.4}$$

For $X_k \in \Omega_e$, use the reflected Euler–Maruyama update

$$X_{k+1} = \left| X_k + \sigma(X_k) Z_k + b(X_k) h \right|. \tag{4.5}$$

Here $\{Z_k\}$ is a sequence of *iid* uniform random variables on $[-\sqrt{3h}, \sqrt{3h}]$.

For $X_k \in \Omega_b$, use the reflected update with probability $\lambda(X_k)$ and jump to 0 with probability $1 - \lambda(X_k)$:

$$X_{k+1} = \begin{cases} \Big| |X_k + \sigma(X_k) Z_k| + b(X_k) h \Big|, & \text{with probability } \lambda(X_k), \\ 0, & \text{with probability } 1 - \lambda(X_k). \end{cases} \tag{4.6}$$

Remark. The nested absolute value in (4.6) is deliberate: reflecting the full increment with a single absolute value leads to an expression for $\lambda(x)$ that need not lie in $[0, 1]$.

The jump probability is

$$\lambda(x) = \frac{(\sigma^2(x) - 2\kappa b(x)) x^2 + 2\kappa \sigma^2(x) x + \sigma^4(x) h}{\left(\frac{\kappa \sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x) \right) x^2 + \sigma^3(x) \kappa \sqrt{3h} + \sigma^4(x) h}. \tag{4.7}$$

This expression is derived in the next subsection. We prove in Section 4.3 that it is a valid probability for sufficiently small h.

Algorithm 2 summarizes the scheme.

Algorithm 2 Approximation of Sticky Diffusion X_T starting from $x \in [0, \infty)$.

Set $X_0 = x$.
for $k = 0$ **to** $N - 1$ **do**
  Generate $Z_k \sim \mathcal{U}([-\sqrt{3h}, \sqrt{3h}])$.
  Calculate boundary threshold $L_k = \sigma(X_k)\sqrt{3h}$.
  if $X_k > L_k$ **then**
    Update X_{k+1} using (4.5).
  else $(X_k \in [0, L_k])$
    Calculate $\lambda(X_k)$ using (4.7).
    Generate $u \sim \mathcal{U}([0, 1])$.
    if $u \le \lambda(X_k)$ **then**
      Update $X_{k+1} = \Big| |X_k + \sigma(X_k) Z_k| + b(X_k) h \Big|$.
    else
      Set $X_{k+1} = 0$.
    end if
  end if
end for

4.2. **How to choose** $\lambda(x)$

As in the SBM case, we choose λ so that the one-step transition operator matches the sticky-diffusion generator in the boundary layer. Expanding $P^h f - f$ gives the condition

$$\left|(\partial_x f)\mathbb{E}_x(\Delta X) + \frac{1}{2}(\partial_{xx} f)\mathbb{E}_x(\Delta X^2) - hb(x)(\partial_x f) - ha(x)(\partial_{xx} f)\right| \leq \epsilon, \tag{4.8}$$

where the local-error analysis below takes $\epsilon = \mathcal{O}(h^2 + hx)$. Taylor expansion of the sticky boundary condition (4.2) about $x = 0$ gives

$$\partial_x f(t,x) = (M(x) + x)\, \partial_{xx} f(t,x) + \mathcal{O}(h + x), \tag{4.9}$$

where

$$M(x) = \frac{\kappa\sigma^2(x)}{\sigma^2(x) - 2\kappa b(x)}.$$

Approximation (4.9) requires $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$; the degenerate case is treated separately in the local-error analysis. Substituting (4.9) into (4.8) and collecting the remainder in ϵ gives

$$(M(x) + x)\left(\mathbb{E}_x(\Delta X) - hb(x)\right) + \frac{1}{2}\mathbb{E}_x(\Delta X^2) - ha(x) = \epsilon. \tag{4.10}$$

We now calculate the first two conditional moments as functions of λ and solve the leading-order equation obtained from (4.10). For $x - \sigma(x)\sqrt{3h} < 0$, the moment calculation gives

$$\begin{aligned}
\mathbb{E}_x(\Delta X_k) &= \frac{\lambda x^2}{2\sqrt{3h}\cdot\sigma(x)} + \frac{\sigma(x)\lambda\sqrt{3h}}{2} + \lambda b(x)h - x + \mathcal{O}(hx + h^2)\,,\\
\mathbb{E}_x(\Delta X_k^2) &= -\frac{\lambda}{\sigma(x)\sqrt{3h}}\cdot x^3 + (1+\lambda)x^2 - \lambda\sigma(x)\sqrt{3h}\cdot x + \lambda\sigma^2(x)\cdot h + \mathcal{O}(hx + h^2)\,.
\end{aligned} \tag{4.11}$$

When $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, substituting (4.11) into (4.10), setting the leading residual to zero, and solving for λ gives (4.7).

4.3. λ **is a probability**

To prove that (4.7) defines a probability for sufficiently small h, we use local coefficient regularity for the boundary-layer expansions and a linear-growth condition for localization and moment control.

Assumption 3 (Local coefficient regularity near the sticky boundary)**.** There exists $r_0 > 0$ such that $b, \sigma \in C^2([0, r_0])$, $\sigma(0) > 0$, and

$$C_{\text{loc}} := \max_{0 \leq j \leq 2}\left(\sup_{x\in[0,r_0]} |\partial_x^j b(x)| + \sup_{x\in[0,r_0]} |\partial_x^j \sigma(x)|\right) < \infty.$$

After decreasing r_0 if necessary, we also assume that

$$\sigma(x) \geq \sigma_* > 0, \qquad x \in [0, r_0].$$

Assumption 4 (Linear growth of the coefficients)**.** There exists $L > 0$ such that

$$|b(x)| + |\sigma(x)| \leq L(1 + x), \qquad x \geq 0.$$

The next lemma localizes the near-boundary region and supplies uniform coefficient bounds there.

Lemma 4.1. *Under Assumptions 3 and 4, there exists $\delta > 0$ and constants $b_1, b_2, \sigma_1, \sigma_2$, independent of x and h, with $\sigma_1 > 0$, such that, for every $h < \delta$,*

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2],$$

whenever

$$x - \sigma(x)\sqrt{3h} < 0 \qquad \text{or} \qquad x - \sigma(x)\sqrt{3h} + b(x)h < 0.$$

Proof. See Supplement. □

Consequently, for $h < \delta$ and $x \in \Omega_b$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2], \qquad x < \sigma(x)\sqrt{3h} \leq \sigma_2\sqrt{3h}.$$

Thus the implicit boundary layer is contained in an $O(\sqrt{h})$ neighborhood of the origin, as in the SBM case.

Lemma 4.2. *Under Assumptions 3 and 4, there exists $\delta > 0$ such that, for every $h < \delta$ and every $x \in \Omega_b$, the jump probability* (4.7) *satisfies*

$$\lambda(x) \in [0, 1].$$

Proof. See Supplement. □

4.4. **Primary convergence results**

Let $u(t, x) = \mathbb{E}[\phi(Y_T) \mid Y_t = x]$ solve the sticky-diffusion backward equation (4.3). Repeating the telescoping argument leading to (3.15), with $t_k = kh$ and $u_k = u(t_k, X_k)$, gives

$$|\mathbb{E}\phi(X_N) - \mathbb{E}\phi(Y_T)| \leq \mathbb{E}\left(\sum_{k=0}^{N-1} |\mathbb{E}\left(u_{k+1} - u_k | X_k\right)| \cdot \mathbf{1}_{X_k \in \Omega_{br}}\right) + \mathbb{E}\left(\sum_{k=0}^{N-1} |\mathbb{E}\left(u_{k+1} - u_k | X_k\right)| \cdot \mathbf{1}_{X_k \notin \Omega_{br}}\right). \tag{4.12}$$

Here $\Omega_{br} = \Omega_b \cup \Omega_r$ with

$$\Omega_r := \{x > \sigma(x)\sqrt{3h};\ x - \sigma(x)\sqrt{3h} + b(x)h \leq 0\}. \tag{4.13}$$

For the error analysis, we enlarge Ω_b to $\Omega_{br} = \Omega_b \cup \Omega_r$ because the two regions have the same form of local error.

The analysis uses the two backward-solution assumptions from the SBM section together with Assumptions 3 and 4. We take $N := T/h \in \mathbb{Z}$.

Theorem 4.3. *(Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in Algorithm 2)*
Under Assumptions 1, 2, 3, and 4, choose $\epsilon = \mathcal{O}(hx + h^2)$ in (4.10). Then there exists a constant $M > 0$, independent of x and h, depending on the relevant derivative bounds of u, the local boundary constants for b, σ, the linear-growth constant of b, σ, and κ, such that for all sufficiently small h,

$$|\mathbb{E}\left(u_{k+1} - u_k \mid X_k = x\right)| \leq \begin{cases} M(h^2 + hx), & x \in \Omega_{br}, \\ Mh^2(1 + x^4), & x \in \Omega_e. \end{cases} \tag{4.14}$$

Thus the boundary-layer local error is $O(h^{3/2})$, while the exterior error is $O(h^2(1 + x^4))$ under the linear-growth condition. Since $N = T/h$, the boundary-layer contribution is $O(h)$ once the following occupation estimate is established.

Lemma 4.4. *For sufficiently small h, there exists a positive constant C, independent of h and initial condition x_0, such that*

$$\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \cdot \mathbf{1}_{X_k \in \Omega_{br}}\right) < C.$$

Its proof reuses most of the construction for Lemma 3.5: after normalizing to $\sigma(0) = 1$, the SBM and sticky-diffusion jump probabilities and transition operators are sufficiently close.

The exterior contribution is controlled by the following fourth-moment estimate, which is compatible with the linear-growth assumption.

Lemma 4.5. *Under Assumptions 3 and 4, suppose that $X_0 = x_0$ deterministically. Then, for $T = Nh$, there exists a constant $C_T > 0$, independent of h and x_0, such that*

$$\sup_{0\le k\le N} \mathbb{E}(1 + X_k^4) \le C_T(1 + x_0^4).$$

Applying these lemmas to (4.12) gives the global error bound for the sticky diffusion, our main convergence theorem.

Theorem 4.6. *(Global Error of $\{X_k\}_{k\in\mathbb{N}}$ in Algorithm 2)*
Let $\phi : [0, \infty) \to \mathbb{R}$ be such that the associated backward solution

$$u(t, x) := \mathbb{E}[\phi(Y_T) \mid Y_t = x]$$

satisfies Assumptions 1 and 2. Suppose also that b and σ satisfy Assumptions 3 and 4, and that $X_0 = Y_0 = x_0$ deterministically. Then there exist constants $\delta > 0$ and $C > 0$, independent of h and x_0, such that for all $h < \delta$,

$$\left|\mathbb{E}\big(\phi(X_N) - \phi(Y_T)\big)\right| \le C(1 + x_0^4)h. \tag{4.15}$$

Proof. Choose $\delta > 0$ sufficiently small that Theorem 4.3 and Lemmas 4.4 and 4.5 apply. Using the local-error estimate in (4.12), together with $\Omega_{br}^c \subseteq \Omega_e$, gives

$$\begin{aligned}
|\mathbb{E}\phi(X_N) - \mathbb{E}\phi(Y_T)| &\le M\mathbb{E}\left[\sum_{k=0}^{N-1}(h^2 + hX_k)\mathbf{1}_{\{X_k\in\Omega_{br}\}}\right] \\
&\quad + Mh^2\sum_{k=0}^{N-1}\mathbb{E}\left[(1 + X_k^4)\mathbf{1}_{\{X_k\in\Omega_e\}}\right] \\
&\le MNh^2 + Mh\,\mathbb{E}\left[\sum_{k=0}^{N-1} X_k\mathbf{1}_{\{X_k\in\Omega_{br}\}}\right] + Mh^2\sum_{k=0}^{N-1}\mathbb{E}(1 + X_k^4) \\
&\le (MT + MC_b)h + MTC_T(1 + x_0^4)h \\
&\le C(1 + x_0^4)h.
\end{aligned}$$

Here C_b and C_T are the constants from Lemmas 4.4 and 4.5, respectively, and we used $Nh = T$ and $1 \le 1 + x_0^4$. The constants δ and C are independent of h and x_0, which proves (4.15). □

4.5. Numerical experiments

Figure 4 empirically validates our convergence results for the choices of b and σ listed in the caption. As in the SBM experiments, we impose a reflecting boundary condition at $x = L$ and compare the stochastic estimates

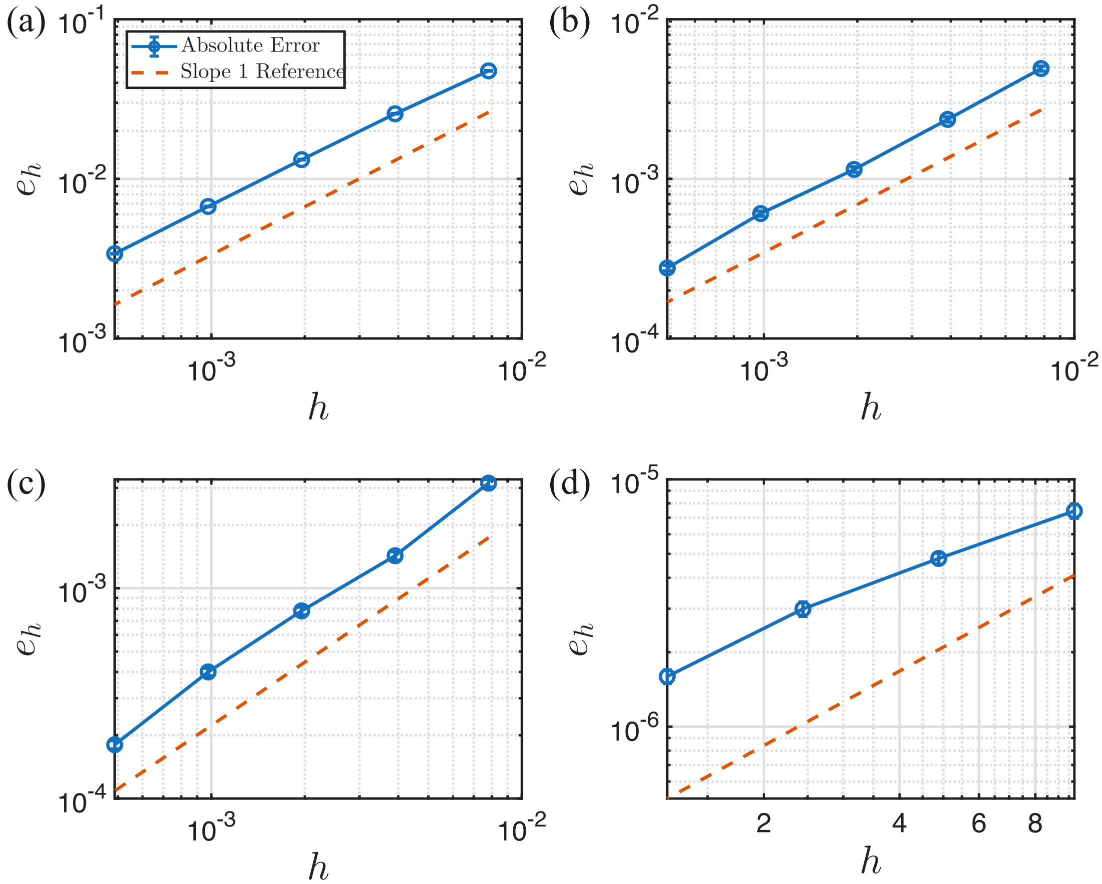

FIGURE 4. Weak error $e_h = |\mathbb{E}[X_T \mid x_0 = 0] - u(T, 0)|$ of Algorithm 2 for different choices of drift, diffusion, and sticky parameter: (a) $b(x) = 2 + \sin(x)$, $\sigma(x) = \frac{1}{2} + x$, $\kappa = 4$; (b) $b(x) = 2 + \sin(x)$, $\sigma(x) = 2 + x$, $\kappa = 1$; (c) $b(x) = -1 + \sin(x)$, $\sigma(x) = 2 + x$, $\kappa = 1$; (d) $b(x) = -2 + \sin(x)$, $\sigma(x) = \frac{1}{2} + x$, $\kappa = 4$. All experiments use $L = 4$, $T = 1$, and $x_0 = 0$. Error bars (commensurate with marker sizes) are ± 1 empirical standard deviation. The respective least-squares fitted slopes are (a) 0.954162 (b) 1.026675, (c) 1.010317, (d) 0.907653 (fitted only through the last 2 data points in this case).

with a "ground truth" obtained by solving the backward equation numerically with a very small finite-difference time step. We again choose $\phi(x) = x$, which does not satisfy the boundary compatibility conditions typically imposed in PDE regularity arguments. Figure 4(a)–(c) used time steps $h \in \{1/128, 1/256, 1/512, 1/1024, 1/2048\}$ and a number of realizations starting at 5×10^7, quadrupled each time h was halved. Figure 4(d) used $h \in \{1/1024, 1/2048, 1/4096, 1/8192\}$, with an initial number of realizations equal to 10^9.

All cases approach $O(h)$ convergence, though it is notable that case (d) with strongly negative drift and weaker diffusion approaches it more slowly, requiring much smaller timesteps to approach a slope of 1. The error in this case, however, is several orders of magnitude smaller than for the others.

4.6. **Key steps in the error analysis**

This subsection records the ingredients for Theorem 4.3 and Lemma 4.4; all proofs are in the Supplement. Relative to SBM, variable diffusion makes the near-boundary set state-dependent, negative drift creates an additional thin reflection strip, and the degenerate boundary relation $b(0) = \sigma^2(0)/(2\kappa)$ requires a separate moment argument. The occupation estimate also requires localizing the enlarged region to an $O(\sqrt{h})$ neighborhood before comparing its transition operator with that of SBM.

4.6.1. *Local error: Theorem 4.3*

The SBM Taylor-expansion template still applies. Outside Ω_{br}, the reflected Euler–Maruyama update has local weak error $O(h^2(1+x^4))$ under linear growth, so the nonstandard analysis is confined to $\Omega_{br} = \Omega_b \cup \Omega_r$. We treat three cases:

(1) $x \in \Omega_b$ and $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$;
(2) $x \in \Omega_r$, which is the additional reflection strip created by negative drift;
(3) $x \in \Omega_b$ and $b(0) = \frac{\sigma^2(0)}{2\kappa}$; then $b(x) > 0$ near the boundary for sufficiently small h, so $\Omega_r = \emptyset$ and $\Omega_{br} = \Omega_b$.

All results below impose Assumptions 3–4; their proofs are in the Supplement.

As in the SBM proof, we first bound $\lambda(x)$ by a multiple of $x/\sqrt{h}+\sqrt{h}$. In particular, $\lambda(x)h^{3/2} = O(hx+h^2)$, which controls the nested-reflection remainders.

Lemma 4.7. *There exist constants $C, \delta > 0$, depending on σ and b, such that, for every $h < \delta$ and $x \in \Omega_b$, the function λ defined in (4.7) satisfies*

$$0 \leq \lambda(x) \leq C\left(\frac{x}{\sqrt{h}} + \sqrt{h}\right).$$

We begin with case 1. The next lemma is the analogue of Lemma 3.2: it replaces $\partial_x u$ by $\partial_{xx} u$ up to a remainder of the required order.

Lemma 4.8. *Let u solve (4.3) with boundary condition (4.2) and assume that $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$. Then, there exists $\delta > 0$ such that for all $h < \delta$ and $x \in \Omega_b$,*

$$\partial_x u(t,x) = (x + M(x))\,\partial_{xx} u(t,x) + Re(x) \tag{4.16}$$

where

$$|Re(x)| \leq C(h+x) = \mathcal{O}(h+x),$$

and C depends on bounds of $\partial_x u, \partial_{xx} u, u^{(3)}$ and on the local bounds of b, σ and their derivatives up to order 2 on the boundary neighborhood $[0, r_0]$, but is independent of x and h.

The next two lemmas give the first and second conditional moments in Ω_b. Relative to SBM, negative drift and the nested absolute value add remainders that Lemma 4.7 bounds by $O(hx+h^2)$.

Lemma 4.9. *There exists $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_b$, we have,*

$$\mathbb{E}_x(\Delta X_k) = \frac{\lambda(x)x^2}{2\sqrt{3h}\cdot\sigma(x)} + \frac{\sigma(x)\lambda(x)\sqrt{3h}}{2} + \lambda(x)b(x)h - x + \mathcal{O}(hx+h^2). \tag{4.17}$$

Lemma 4.10. *There exists $\delta > 0$ and $C > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_b$, we have,*

$$\mathbb{E}_x(\Delta X_k^2) = -\frac{\lambda(x)}{\sigma(x)\sqrt{3h}}\cdot x^3 + (1+\lambda(x))x^2 - \lambda(x)\sigma(x)\sqrt{3h}\cdot x + \lambda(x)\sigma^2(x)\cdot h + R_2(x),$$

where

$$|R_2(x)| \leq C\cdot(hx+h^2) = \mathcal{O}(hx+h^2).$$

Combining these moment formulas yields the leading-order cancellation built into the choice of $\lambda(x)$.

Lemma 4.11. *If $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, then there exists $\delta > 0$ such that, for every $h < \delta$ and $x \in \Omega_b$,*

$$2\left(\mathbb{E}_x(\Delta X_k) - hb(x)\right) \cdot (x + M(x)) + \mathbb{E}_x(\Delta X_k^2) = \sigma^2(x)h + R_3(x) \tag{4.18}$$

where $|R_3(x)| \leq K_3 \cdot (hx + h^2) = \mathcal{O}(hx + h^2)$ for some K_3 independent of h but depending on the boundary-layer constants and the nondegeneracy constant from $b(0) \neq \sigma^2(0)/(2\kappa)$.

The local Taylor estimate also requires bounds on the first and third moments. Although jumps to 0 are $O(\sqrt{h})$, the preceding cancellation lowers the first moment in Ω_b to order h.

Lemma 4.12. *There exists $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_b$,*

$$|\mathbb{E}_x(\Delta X_k)| \leq C(h + hx + h^2), \tag{4.19}$$

where $C > 0$ is independent of x and h and depends only on κ, the local boundary bounds for b, σ, and the nondegeneracy constant coming from $b(0) \neq \sigma^2(0)/(2\kappa)$.

Lemma 4.13. *There exists $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_{br}$,*

$$\left|\mathbb{E}_x(\Delta X_k^3)\right| \leq K_4(hx + h^2), \tag{4.20}$$

where K_4 is a fixed constant dependent on drift, diffusion, and κ.

The SBM Taylor argument, with Lemmas 4.8 and 4.11 providing the modified cancellation, now gives the case-1 local error.

Theorem 4.14. *(Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in algorithm 2, situation 1)*
Given $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, there exists $\delta > 0$ and $C \geq 0$ depending on the relevant bounds of derivatives of u and on the local boundary constants for b, σ such that $\forall h < \delta$ and $\forall x \in \Omega_b$,

$$|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| \leq C(h^2 + hx) = \mathcal{O}(hx + h^2) \tag{4.21}$$

For case 2, $x \in \Omega_r$, the scheme uses a reflected Euler–Maruyama step with no jump to 0. The next lemma shows that its first moment still approximates $b(x)h$ at the required order.

Lemma 4.15. *There exists $\delta > 0$ and a constant C, which depends only on local boundary constants for b, σ coming from Lemma 4.1 such that $\forall h < \delta$ and $\forall x \in \Omega_r$,*

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| \leq C(hx + h^2) = \mathcal{O}(hx + h^2).$$

The corresponding second moment is $\sigma^2(x)h + O(hx + h^2)$, so the same Taylor argument yields the case-2 local error.

Theorem 4.16. *(Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in algorithm 2, situation 2)*
Given $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, there exists $\delta > 0$ and $C \geq 0$ depending on the relevant derivative bounds of u and on the local boundary constants for b and σ such that $\forall h < \delta$ and $\forall x \in \Omega_r$,

$$|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| \leq C(h^2 + hx) = \mathcal{O}(hx + h^2). \tag{4.22}$$

Cases 1 and 2 cover $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$. When equality holds, $M(x)$ need not be locally bounded, so case 3 instead uses a direct first-moment estimate in Ω_b.

Lemma 4.17. *When $b(0) = \frac{\sigma^2(0)}{2\kappa}$, there exist $\delta > 0$ and a constant C which depends only on the boundary-layer bounds, such that $\forall h < \delta$ and $\forall x \in \Omega_b$,*

$$|\mathbb{E}_x\left(\Delta X_k\right) - hb(x)| \leq C(hx + h^2) = \mathcal{O}(hx + h^2).$$

Because $b(0) = \sigma^2(0)/(2\kappa) > 0$, continuity rules out Ω_r for sufficiently small h. The preceding moment estimate therefore yields the case-3 local error in $\Omega_b = \Omega_{br}$.

Theorem 4.18. *(Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in algorithm 2, situation 3)*
If $b(0) = \frac{\sigma^2(0)}{2\kappa}$, there exist $\delta > 0$ and $C \geq 0$, independent of x and h, depending only on the relevant bounds of derivatives of u and on the local boundary constants for b, σ such that, for every $h < \delta$ and $x \in \Omega_{br}$,

$$|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| \leq C(h^2 + hx) = \mathcal{O}(hx + h^2). \tag{4.23}$$

4.6.2. *Bounding the number of steps: Lemma 4.4*

To enable comparison with the SBM construction, we first normalize the boundary diffusion coefficient. Let $\sigma_0 := \sigma(0) > 0$ and define the rescaled process

$$\hat{X}_t := \frac{X_t}{\sigma_0}.$$

Then $\hat{X}$ is again a sticky diffusion of the same form, with coefficients

$$\hat{b}(y) := \frac{b(\sigma_0 y)}{\sigma_0}, \qquad \hat{\sigma}(y) := \frac{\sigma(\sigma_0 y)}{\sigma_0},$$

and sticky parameter $\hat{\kappa} = \kappa/\sigma_0$. In particular, $\hat{\sigma}(0) = 1$. Therefore it is enough to prove the desired estimate in the normalized case $\sigma(0) = 1$, and we adopt this normalization below.

Our hope is to recycle our calculation for the global error analysis done in the proof of Theorem 3.6 to show similar theorems for the scheme with drift.

The first thing we notice is that when $\sigma(x) = 1$, then the jump probability λ for the general diffusion, (4.7), is

$$\lambda(x) = \frac{(1 - 2\kappa b(x))x^2 + 2\kappa x + h}{\left(\frac{\kappa}{\sqrt{3h}} + 1 - 2\kappa b(x)\right) x^2 + h + \kappa\sqrt{3h}},$$

which differs from the jump probability for the SBM, (3.3), only in terms involving $b(x)$. Since these terms multiply x^2, which is small in the boundary layer, and since if $\sigma(0) = 1$ then $\sigma(x) \approx 1$ in the boundary layer, we expect that these two jump probabilities will be not too different. The next lemma shows that the difference between the two is $\mathcal{O}(\sqrt{h})$.

Lemma 4.19. *Denote $\tilde{\lambda}(\cdot)$ as the jump probability for SBM,* (3.3)*, and $\lambda(\cdot)$ as the jump probability for the general diffusion,* (4.7)*. If $\sigma(0) = 1$, then*

$$\left|\lambda(x) - \tilde{\lambda}(x)\right| \leq K_{b,\sigma}\sqrt{h}, \quad \forall x \in [0, \sqrt{3h}],$$

where $K_{b,\sigma} > 0$ is a constant depending on the drift term $b(\cdot)$, diffusion $\sigma(\cdot)$, and sticky parameter κ.

These results may be used to show Lemma 4.4, which is analogous to Lemma 3.5, and is the final step in proving the order of global error. The proof of Lemma 4.4 is in the Supplement.

5. Conclusion

This paper presented and analyzed a numerical scheme for simulating one-dimensional sticky diffusions. The scheme combines a reflected Euler-Maruyama increment, with a carefully designed jump probability to the sticky point, to handle the challenge of moving both in continuous space, and to a lower-dimensional boundary. Our scheme achieves a weak local error of order 3/2 within the boundary layer and a weak global error of order 1.

We provide rigorous theoretical analysis of our schemes, including detailed proofs of local and global error bounds. Numerical experiments corroborate our theoretical findings, showing empirical evidence of first-order global convergence for both sticky Brownian motion and sticky diffusion processes under various conditions.

A natural extension of our work would be to develop schemes for multidimensional sticky diffusions. A novel feature of such diffusions, compared with the one-dimensional case, is that they can have nontrivial dynamics along their sticky boundaries, which can be different from the dynamics in the interior. We are optimistic that the method we have outlined can be adapted to such diffusions, because it provides a clear guideline for choosing how to switch between different types of dynamics (i.e. on sets of different dimensions), which are otherwise standard to simulate. A challenge however will be ensuring that the switching probabilities are indeed probabilities, as we have found this nontrivial to ensure even for a one-dimensional diffusion. Further extensions could include adapting these methods to processes on manifolds [11], and considering multiple intersecting sticky boundaries which form a stratification, as outlined in [21, 22].

Even for a one-dimensional diffusion there is further theoretical development possible. One could investigate higher-order schemes, which might achieve even better convergence rates, or could ask how to solve general parabolic-type PDEs with sticky boundary conditions [31, 45]. One could also ask for *worse* schemes, which achieve only $O(h^{1/2})$ convergence – perhaps there is a wider scope of jump probabilities λ which allow for worse convergence, but are easier to either derive, or to implement. Such analysis could perhaps be applied to Monte-Carlo schemes which perform local moves to sample from a stationary distribution supported on manifolds of different dimensions, and show their convergence to a particular sticky diffusion [21]. Finally, it would be useful to perform a theoretical analysis of the long-time behavior of the numerical schemes and their ability to capture stationary distributions accurately.

In conclusion, this work contributes to the growing toolkit for numerical simulation of stochastic processes with complex boundary conditions, offering both practical algorithms and theoretical guarantees for simulating certain sticky processes. As the study of such processes continues to grow in importance across various disciplines, we believe these methods will prove valuable tools for researchers and practitioners alike.

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Appendix A. Supplement

A.1. Additional proofs from Section 3

Verification of (3.23) **in the proof of Lemma 3.5**

As a reminder, we wish to show that there exist constants $C_1(s,M) > 0$ and $\delta > 0$, independent of x and h, such that for all $h < \delta$,

$$Pe^{w(x)} \leq e^{w(x)} \left(1 - f_h(x) + C_1 h\right),$$

where f_h satisfies

$$f_h(x) \geq x\, \mathbf{1}_{\{x\in[0,\sqrt{3h}]\}}.$$

We would like to calculate $Pe^{w(x)} - e^{w(x)}$ for x in three cases: $[0, \frac{\sqrt{3h}}{2}]$, $[\frac{\sqrt{3h}}{2}, \sqrt{3h}]$ and $[\sqrt{3h}, +\infty)$.

(1) For $x \in [0, \frac{\sqrt{3h}}{2}]$, we compute

$$\begin{aligned}
Pe^{w(x)} &= \frac{\lambda(x)}{2}e^M + \frac{\lambda(x)}{\sqrt{3h}} \cdot \int_{\frac{\sqrt{3h}}{2}}^{\sqrt{3h}-x} e^{M-s\left(z-\frac{\sqrt{3h}}{2}\right)}\, dz + \frac{\lambda(x)}{2\sqrt{3h}} \cdot \int_{\sqrt{3h}-x}^{\sqrt{3h}+x} e^{M-s\left(z-\frac{\sqrt{3h}}{2}\right)}\, dz \\
&\quad + e^M(1-\lambda(x)) \\
&= \lambda(x)\left(\frac{1}{2}e^M + e^M \cdot \frac{1}{2\sqrt{3h}s} \cdot \left(2 - e^{-s\left(\frac{\sqrt{3h}}{2}-x\right)} - e^{-s\left(\frac{\sqrt{3h}}{2}+x\right)}\right)\right) + e^M(1-\lambda(x)).
\end{aligned}$$

In the first line, the first term corresponds to reflected jumps that land in the flat part of w, the two integral terms correspond to reflected jumps that land in the sloping part of w (split according to whether the reflection map has one or two preimages), and the last term corresponds to the jump directly to 0.

Now we simplify this expression by Taylor expansion of exponentials with a remainder of order $\mathcal{O}(h)$:

$$\begin{aligned}
&Pe^{w(x)} - e^{w(x)} \\
&= e^M \cdot \frac{\lambda(x)}{2\sqrt{3h}s} \cdot \left(2 - e^{-s\left(\frac{\sqrt{3h}}{2}-x\right)} - e^{-s\left(\frac{\sqrt{3h}}{2}+x\right)}\right) - e^M \cdot \frac{\lambda(x)}{2} \\
&= e^M \cdot \frac{\lambda(x)}{2\sqrt{3h}s} \cdot \left(\sqrt{3h}s - \frac{1}{2}s^2\left(\frac{3h}{2} + 2x^2\right) + \mathcal{O}(h^{3/2})\right) - e^M \cdot \frac{\lambda(x)}{2} \\
&= e^M \cdot \frac{\lambda(x)}{2\sqrt{3h}s} \cdot \left(-\frac{1}{2}s^2\left(\frac{3h}{2} + 2x^2\right) + \mathcal{O}(h^{3/2})\right) \\
&= e^M \cdot \left(-\frac{s}{4} \cdot \frac{\lambda(x)}{2}\sqrt{3h} - s \cdot \frac{\lambda(x)x^2}{2\sqrt{3h}} + \mathcal{O}(h)\right).
\end{aligned}$$

In other words, we have shown that, for $x \in [0, \sqrt{3h}/2]$,

$$Pe^{w(x)} \leq e^{w(x)}\left(1 - f_h(x) + \mathcal{O}(h)\right)$$

where we define

$$f_h(x) := \frac{s}{4} \cdot \frac{\lambda(x)}{2}\sqrt{3h} + s \cdot \frac{\lambda(x) \cdot x^2}{2\sqrt{3h}} \qquad \text{for } x \in [0, \frac{\sqrt{3h}}{2}]. \tag{A.1}$$

Recall the expression for $\mathbb{E}_x(\Delta X)$, (3.9) and substitute the expression for $\lambda(x)$, (3.3),

$$\begin{aligned}
\mathbb{E}_x(\Delta X_k) &= -x + \frac{\lambda(x)\sqrt{3h}}{2} + \frac{\lambda(x)x^2}{2\sqrt{3h}} \\
&= \frac{1}{2\sqrt{3h}} \cdot \frac{\left(x - \sqrt{3h}\right)^2 (x^2 + h)}{\left(\frac{\kappa}{\sqrt{3h}} + 1\right)x^2 + h + \kappa\sqrt{3h}} \\
&\geq 0.
\end{aligned}$$

Hence,

$$f_h(x) - x > \mathbb{E}_x(\Delta X) > 0, \quad \forall x \in [0, \frac{\sqrt{3h}}{2}].$$

(2) For $x \in [\frac{\sqrt{3h}}{2}, \sqrt{3h}]$, the scheme does the same jump, but the expression for $e^{w(x)}$ is different, which makes the calculation much more complex. It can be checked that

$$\begin{aligned}
Pe^{w(x)} &= \lambda(x)\left(e^M \cdot \frac{\sqrt{3h} - x}{\sqrt{3h}} + e^M \cdot \frac{\frac{\sqrt{3h}}{2} - (\sqrt{3h} - x)}{2\sqrt{3h}}\right) \\
&\quad + \frac{\lambda(x)}{2\sqrt{3h}} \cdot \int_{\frac{\sqrt{3h}}{2}}^{\sqrt{3h}+x} e^{M - s\left(z - \frac{\sqrt{3h}}{2}\right)}\, dz + (1 - \lambda(x))e^M \\
&= \lambda(x)e^M \cdot \frac{3\sqrt{3h} - 2sx + 2 - 2e^{-s\left(\frac{\sqrt{3h}}{2}+x\right)}}{4\sqrt{3h}s} + (1 - \lambda(x))e^M.
\end{aligned}$$

Since $e^{w(x)} = e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)}$,

$$Pe^{w(x)} - e^{w(x)} = A + B$$

where

$$\begin{aligned}
A &= \lambda(x)e^M \cdot \frac{3\sqrt{3h} - 2sx + 2 - 2e^{-s\left(\frac{\sqrt{3h}}{2}+x\right)}}{4\sqrt{3h}s} - \lambda(x) \cdot e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)} \\
B &= (1-\lambda(x)) \cdot e^M - (1-\lambda(x)) \cdot e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)}.
\end{aligned}$$

Simplifying A, B by expanding the exponential, we have

$$\begin{aligned}
A &= \frac{\lambda(x)}{4\sqrt{3h}s} \cdot e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)} \cdot \left((3\sqrt{3h}s - 2sx + 2)e^{s\left(x-\frac{\sqrt{3h}}{2}\right)} - 2e^{-s\sqrt{3h}} - 4\sqrt{3h}s\right) \\
&= e^{w(x)} \cdot \frac{\lambda(x)}{4\sqrt{3h}s} \cdot \left(3\sqrt{3h}s^2x - \frac{9}{4}s^2 \cdot 3h - s^2x^2 + \mathcal{O}(h^{3/2})\right) \\
&= \frac{\lambda(x)}{4}e^{w(x)}\left(3sx - \frac{9}{4}\sqrt{3h}s - \frac{sx^2}{\sqrt{3h}} + \mathcal{O}(h)\right).
\end{aligned} \tag{A.2}$$

Meanwhile,

$$\begin{aligned}
B &= e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)} \cdot (1-\lambda(x))\left(e^{s\left(x-\frac{\sqrt{3h}}{2}\right)} - 1\right) \\
&= e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)} \cdot (1-\lambda(x))\left(s\left(x - \frac{\sqrt{3h}}{2}\right) + \mathcal{O}(h)\right).
\end{aligned} \tag{A.3}$$

Combining (A.2), (A.3) together, we have

$$Pe^{w(x)} - e^{w(x)} = A + B = -e^{w(x)}\left(f_h(x) + \mathcal{O}(h)\right)$$

where we define

$$f_h(x) := \frac{s\lambda(x)}{16}\sqrt{3h} + \frac{s\lambda(x)x^2}{4\sqrt{3h}} + \frac{s\lambda(x)x}{4} + \frac{s\sqrt{3h}}{2} - sx \quad \text{for } x \in [\frac{\sqrt{3h}}{2}, \sqrt{3h}]. \tag{A.4}$$

It can be checked by derivative that $f_h(x)$ is decreasing for $x \in [\frac{\sqrt{3h}}{2}, \sqrt{3h}]$, and $f_h(\sqrt{3h}) = \frac{s}{16}\sqrt{3h} > \sqrt{3h}$, so we must have that

$$f_h(x) - x > 0, \quad \forall x \in [\frac{\sqrt{3h}}{2}, \sqrt{3h}].$$

(3) For $x \in (\sqrt{3h}, +\infty)$: in this region, the scheme does a standard EM jump. Using the fact that $\mathbb{E}_x(\Delta X) = 0$, and then using the fact that w is a piecewise linear function with a constant piece near the boundary,

so it smaller than its linear part: $w(x+\Delta X) \le M - s\left((x+\Delta X) - \frac{\sqrt{3h}}{2}\right)$, we have

$$\begin{aligned}
Pe^{w(x)} &= \mathbb{E}_x\left(e^{w(x+\Delta X)}\right)\\
&\le \mathbb{E}_x\left(e^{w(x)-s\cdot\Delta X}\right)\\
&= e^{w(x)}\cdot(1 - s\cdot\mathbb{E}_x(\Delta X) + \mathcal{O}(h))\\
&= e^{w(x)}\cdot(1 + 0\cdot(-s) + \mathcal{O}(h))\\
&= e^{w(x)}\left(1 - f_h(x) + \mathcal{O}(h)\right)
\end{aligned}$$

where

$$f_h(x) := 0, \qquad \text{for } x \in (\sqrt{3h}, +\infty). \tag{A.5}$$

So far, we have shown that for $x \in [0,+\infty)$,

$$Pe^{w(x)} \le e^{w(x)}\left(1 - f_h(x) + \mathcal{O}(h)\right) \tag{A.6}$$

where $f_h(x) \ge x \cdot \mathbf{1}_{x\in[0,\sqrt{3h}]}$ is defined according to (A.1),(A.4),(A.5), and the bounding constant of the big-$\mathcal{O}$ term depends on s. In each of the three cases above, the remainder term $\mathcal{O}(h)$ is uniform in x on the corresponding interval. Therefore, after taking the maximum of the three remainder constants and the minimum of the corresponding smallness thresholds, there exist constants $C_1(s,M) > 0$ and $\delta > 0$, independent of x and h, such that for all $h < \delta$,

$$Pe^{w(x)} \le e^{w(x)}\big(1 - f_h(x) + C_1 h\big), \qquad x \ge 0.$$

Together with the bounds established above,

$$f_h(x) \ge x\,\mathbf{1}_{\{x\in[0,\sqrt{3h}]\}},$$

which proves (3.23).

A.2. **Additional proofs from Section 4**

Lemma 4.1. There exists $\delta > 0$ and constants $b_1, b_2, \sigma_1, \sigma_2$ such that, for every $h < \delta$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2],$$

whenever

$$x - \sigma(x)\sqrt{3h} < 0 \qquad \text{or} \qquad x - \sigma(x)\sqrt{3h} + b(x)h < 0.$$

Proof. We first show that the boundary layer is contained in a fixed compact neighborhood of 0 for all sufficiently small h. By Assumption 4, there exists $L > 0$ such that

$$|b(x)| + |\sigma(x)| \le L(1+x), \qquad x \ge 0.$$

If $x - \sigma(x)\sqrt{3h} < 0$, then

$$x < \sigma(x)\sqrt{3h} \le |\sigma(x)|\sqrt{3h} \le L(1+x)\sqrt{3h}.$$

If $x - \sigma(x)\sqrt{3h} + b(x)h < 0$, then

$$x < \sigma(x)\sqrt{3h} - b(x)h \le |\sigma(x)|\sqrt{3h} + |b(x)|h \le L(1+x)(\sqrt{3h} + h).$$

Hence, after choosing $\delta_0 > 0$ small enough so that

$$L(\sqrt{3\delta_0} + \delta_0) \leq \frac{1}{2},$$

both cases imply

$$x \leq 2L(\sqrt{3h} + h), \qquad h < \delta_0.$$

In particular, by decreasing δ_0 if necessary, every such x lies in the interval $[0, r_0]$ from Assumption 3. Moreover,

$$|\partial_x b(y)| + |\partial_x \sigma(y)| \leq C_{\text{loc}}, \qquad y \in [0, r_0].$$

Now we sharpen this localization using the local boundedness of $\partial_x b$ and $\partial_x \sigma$ on $[0, r_0]$. Denote

$$\sigma_0 := \sigma(0), \qquad b_0 := b(0),$$

If $x - \sigma(x)\sqrt{3h} + b(x)h < 0$, then by the mean value theorem,

$$\begin{aligned} x &< \sigma(x)\sqrt{3h} - b(x)h \\ &= (\sigma_0 + x\partial_x\sigma(z_x))\sqrt{3h} - (b_0 + x\partial_x b(y_x))\,h \end{aligned} \tag{A.7}$$

for some $y_x, z_x \in [0, x]$. Therefore

$$x\left(1 - \partial_x\sigma(z_x)\sqrt{3h} + \partial_x b(y_x)h\right) < \sigma_0\sqrt{3h} - b_0 h.$$

After decreasing δ_0 if necessary, we have

$$1 - \partial_x\sigma(z_x)\sqrt{3h} + \partial_x b(y_x)h \geq \frac{1}{2}, \qquad h < \delta_0,$$

and hence

$$x < \sigma_0\sqrt{3h} + C_1 h,$$

where C_1 depends only on b_0, σ_0 and the local bound C_{loc}.
Similarly, if $x - \sigma(x)\sqrt{3h} < 0$, then

$$x < \sigma(x)\sqrt{3h} = (\sigma_0 + x\partial_x\sigma(z_x))\sqrt{3h}, \tag{A.8}$$

so

$$x\left(1 - \partial_x\sigma(z_x)\sqrt{3h}\right) < \sigma_0\sqrt{3h}.$$

After decreasing δ_0 if necessary,

$$1 - \partial_x\sigma(z_x)\sqrt{3h} \geq \frac{1}{2},$$

and therefore

$$x < \sigma_0\sqrt{3h} + C_2 h,$$

where C_2 depends only on σ_0 and C_{loc}. Let $C := C_1 \vee C_2$. We have shown that for every $h < \delta_0$,

$$\left\{x \geq 0 :\; x - \sigma(x)\sqrt{3h} < 0 \text{ or } x - \sigma(x)\sqrt{3h} + b(x)h < 0\right\} \subseteq [0, \sigma_0\sqrt{3h} + Ch].$$

Now choose $\delta \in (0, \delta_0]$ small enough so that

$$B_\delta := [0, \sigma_0\sqrt{3\delta} + C\delta] \subseteq [0, r_0].$$

Since b and σ are continuous on B_δ, define

$$b_1 := \min_{x\in B_\delta} b(x), \qquad b_2 := \max_{x\in B_\delta} b(x),$$

and

$$\sigma_1 := \min_{x\in B_\delta} \sigma(x), \qquad \sigma_2 := \max_{x\in B_\delta} \sigma(x).$$

By Assumption 3, after decreasing δ if necessary, $\sigma_1 > 0$. Therefore, for every $h < \delta$ and every x satisfying either boundary-layer inequality,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2],$$

as claimed. $\square$

As a side remark, σ_i and b_i are different from $\sigma_0 = \sigma(0)$ and $b_0 = b(0)$ up to a term of order $\sqrt{h}$ because we have shown the boundary layer of our scheme is included in set B_δ. Lemma 4.1 is useful in the sense that it holds when we want to do error analysis for our scheme whenever we do not make a standard EM jump.

Lemma 4.2. There exists $\delta > 0$ such that $\forall h < \delta$, $\forall x$ such that $x - \sigma(x)\sqrt{3h} < 0$,

$$\lambda(x) \in [0, 1].$$

Proof. By Lemma 4.1, there exist $\delta_1 > 0$ and constants $b_1, b_2, \sigma_1, \sigma_2$, depending only on the local boundary constants and the linear-growth constant of b, σ, such that for every $h < \delta_1$ and every x with $x - \sigma(x)\sqrt{3h} < 0$, we have

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].$$

Hence

$$x < \sigma(x)\sqrt{3h} \le \sigma_2\sqrt{3h}.$$

Therefore it suffices to prove the following algebraic statement: for all $(x, b, \sigma) \in [0, \sigma_2\sqrt{3h}] \times [b_1, b_2] \times [\sigma_1, \sigma_2]$, the quantity $\Lambda(x, b, \sigma)$ defined below (A.10) satisfies

$$0 \le \Lambda(x, b, \sigma) \le 1.$$

If $b_2 \le 0$, choose $\delta_2 = 1$; if $b_2 > 0$, δ_2 is chosen in such a way that the following inequalities hold for all $h < \delta_2$,

$$\frac{\kappa}{\sqrt{3h}} > \frac{2\kappa b_2\sigma_2}{\sigma_1^2}, \qquad \sqrt{3h} < \frac{\sigma_2}{2b_2},$$

which implies that for all $b \in [b_1, b_2]$ and $\sigma \in [\sigma_1, \sigma_2]$

$$\begin{aligned} \frac{\kappa}{\sqrt{3h}} &> \frac{(2\kappa b - \sigma^2)\sigma_2}{\sigma^2}, \\ \frac{\kappa\sigma}{\sqrt{3h}} &> -(\sigma^2 - 2\kappa b). \end{aligned} \tag{A.9}$$

Consider, $\forall h < \min\{\delta_1, \delta_2\}$,

$$\Lambda(x, b, \sigma) = \frac{(\sigma^2 - 2\kappa b)x^2 + 2\kappa\sigma^2 x + \sigma^4 h}{\left(\frac{\kappa\sigma}{\sqrt{3h}} + \sigma^2 - 2\kappa b\right)x^2 + \sigma^3\kappa\sqrt{3h} + \sigma^4 h}, \tag{A.10}$$

and

$$1-\Lambda(x,b,\sigma)=\frac{\frac{\kappa\sigma}{\sqrt{3h}}\left(x-\sigma\sqrt{3h}\right)^2}{\left(\frac{\kappa\sigma}{\sqrt{3h}}+\sigma^2-2\kappa b\right)x^2+\sigma^3\kappa\sqrt{3h}+\sigma^4 h}. \tag{A.11}$$

It suffices to conclude the lemma if we can show $\forall(x,b,\sigma)\in[0,\sigma_2\sqrt{3h}]\times[b_1,b_2]\times[\sigma_1,\sigma_2]$,

$$\Lambda(x,b,\sigma)\geq 0, \quad \text{and} \quad 1-\Lambda(x,b,\sigma)\geq 0. \tag{A.12}$$

We consider three cases: (i) $b<\frac{\sigma^2}{2\kappa}$ (ii) $b>\frac{\sigma^2}{2\kappa}$; (iii) $b=\frac{\sigma^2}{2\kappa}$.

(1) For $b<\frac{\sigma^2}{2\kappa}$, we notice that both numerator (denote as $N_1(x,b)$) and denominator (denote as $D(x,b)$) of $\Lambda(x,b)$ are strictly positive. Besides, the numerator of $1-\Lambda(x,b)$ is a square and its denominator is identical to $D(x,b)$, so it is strictly positive. (A.12) is verified.

(2) For $b>\frac{\sigma^2}{2\kappa}$, we have $2\kappa b-\sigma^2>0$, and in particular $b_2>0$. From the first inequality in (A.9), for every $x\in[0,\sigma_2\sqrt{3h}]$,

$$(2\kappa b-\sigma^2)x\leq(2\kappa b-\sigma^2)\sigma_2\sqrt{3h}<\kappa\sigma^2.$$

Hence

$$2\kappa\sigma^2-(2\kappa b-\sigma^2)x>\kappa\sigma^2>0.$$

Using this, the numerator of Λ satisfies

$$\begin{aligned}N_1(x,b,\sigma)&=(\sigma^2-2\kappa b)x^2+2\kappa\sigma^2x+\sigma^4h\\&=\sigma^4h+x\big(2\kappa\sigma^2-(2\kappa b-\sigma^2)x\big)\\&>\sigma^4h+\kappa\sigma^2x\\&>0.\end{aligned}$$

For the denominator, by the second inequality in (A.9),

$$\frac{\kappa\sigma}{\sqrt{3h}}+\sigma^2-2\kappa b>0.$$

Therefore

$$\begin{aligned}D(x,b,\sigma)&=\left(\frac{\kappa\sigma}{\sqrt{3h}}+\sigma^2-2\kappa b\right)x^2+\sigma^3\kappa\sqrt{3h}+\sigma^4h\\&\geq\sigma^3\kappa\sqrt{3h}+\sigma^4h\\&>0.\end{aligned}$$

Thus $\Lambda(x,b,\sigma)\geq 0$ on $[0,\sigma_2\sqrt{3h}]\times[b_1,b_2]\times[\sigma_1,\sigma_2]$. We also have $1-\Lambda(x,b,\sigma)\geq 0$. Hence (A.12) holds in this case.

(3) For $b=\frac{\sigma^2}{2\kappa}$, by simple algebra,

$$\Lambda(x,b,\sigma)=\frac{2\sigma^2x+2b\sigma^2h}{\frac{\sigma x^2}{\sqrt{3h}}+\sigma^3\sqrt{3h}+2b\sigma^2h}\geq 0,$$

$$1-\Lambda(x,b,\sigma)=\frac{\frac{\sigma}{\sqrt{3h}}\left(x-\sigma\sqrt{3h}\right)^2}{\frac{\sigma x^2}{\sqrt{3h}}+\sigma^3\sqrt{3h}+2b\sigma^2h}\geq 0.$$

Hence (A.12) is verified in this case.

$\square$

Lemma 4.7. There exists constant $C, \delta > 0$ dependent on $\sigma(x)$ and $b(x)$ such that $\forall h < \delta$ and $\forall x \in [0, \sigma_2\sqrt{3h}]$ such that $x - \sigma(x)\sqrt{3h} < 0$,

$$0 \le \lambda(x) \le C \cdot \left(\frac{x}{\sqrt{h}} + \sqrt{h} \right).$$

Proof. By Lemma 4.1, after decreasing δ if necessary, there exist constants $b_1, b_2, \sigma_1, \sigma_2$, independent of x and h, such that whenever $h < \delta$ and $x - \sigma(x)\sqrt{3h} < 0$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2],$$

and hence

$$0 \le x < \sigma(x)\sqrt{3h} \le \sigma_2\sqrt{3h}.$$

Let $B_M := |b_1| \vee |b_2|$, by Lemma 4.2, after decreasing δ if necessary, we already know that $\lambda(x) \in [0, 1]$. It remains to prove the sharper upper bound. Write

$$\lambda(x) = \frac{N_h(x)}{D_h(x)},$$

where

$$N_h(x) := (\sigma^2(x) - 2\kappa b(x))x^2 + 2\kappa\sigma^2(x)x + \sigma^4(x)h,$$

and

$$D_h(x) := \left(\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x) \right) x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^4(x)h.$$

Decrease δ once more so that, for all $h < \delta$,

$$\frac{\kappa\sigma_1}{\sqrt{3h}} \ge 2\kappa B_M.$$

Then, for all x under consideration,

$$\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x) \ge \frac{\kappa\sigma_1}{\sqrt{3h}} - 2\kappa B_M \ge 0.$$

Therefore

$$D_h(x) \ge \sigma^3(x)\kappa\sqrt{3h} \ge \sigma_1^3\kappa\sqrt{3h}.$$

On the other hand,

$$|N_h(x)| \le (\sigma_2^2 + 2\kappa B_M)x^2 + 2\kappa\sigma_2^2 x + \sigma_2^4 h.$$

Since $x \le \sigma_2\sqrt{3h}$, we have $x^2 \le 3\sigma_2^2 h$. Hence

$$\begin{aligned} \lambda(x) &\le \frac{|N_h(x)|}{D_h(x)} \\ &\le \frac{(\sigma_2^2 + 2\kappa B_M)\sigma_2^2 \cdot 3h + 2\kappa\sigma_2^2 x + \sigma_2^4 h}{\sigma_1^3\kappa\sqrt{3h}} \\ &= \frac{2\sigma_2^2}{\sqrt{3}\sigma_1^3} \cdot \frac{x}{\sqrt{h}} + \frac{4\sigma_2^4 + 6\kappa B_M\sigma_2^2}{\sqrt{3}\sigma_1^3\kappa} \cdot \sqrt{h}. \end{aligned} \tag{A.13}$$

Choosing

$$C := \max\left\{ \frac{2\sigma_2^2}{\sqrt{3}\sigma_1^3}, \frac{4\sigma_2^4 + 6\kappa B_M\sigma_2^2}{\sqrt{3}\sigma_1^3\kappa} \right\}$$

completes the proof. □

Lemma 4.8. Let u be the solution of (4.3) satisfying corresponding boundary condition (4.2) and assume that $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$. Then, there exists δ small such that for all $h < \delta$ and $x \in \Omega_b$,

$$\partial_x u(t,x) = (x + M(x))\, \partial_{xx} u(t,x) + Re(x)$$

where

$$|Re(x)| \le C(h+x) = \mathcal{O}(h+x),$$

and C depends on bounds of $\partial_x u, \partial_{xx} u, u^{(3)}$ and on the local bounds of b, σ and their derivatives up to order 2 on the boundary neighborhood $[0, r_0]$, but is independent of x and h.

Proof. Similar to proof of Lemma 3.2, apply Taylor expansion with remainder to (4.2) centered at x. By Lemma 4.1, after decreasing δ if necessary, every $x \in \Omega_b$ lies in $B_\delta \subset [0, r_0]$. Hence all intermediate points in the Taylor expansions below also lie in $[0, r_0]$, where the local coefficient bounds from Assumption 3 apply. We have

$$\begin{aligned}
\sigma^2(0)\partial_x u(t,0) &= \sigma^2(x)\partial_x u(t,x) - x\sigma^2(x)\partial_{xx} u(t,x) - 2x\sigma(x)\sigma'(x)\partial_x u(t,x) \\
&\quad + \frac{x^2}{2}\partial_{xx}\left(\sigma^2(y_{1,x})\partial_x u(t, y_{1,x})\right) \\
&= \sigma^2(x)\partial_x u(t,x) - x\sigma^2(x)\partial_{xx} u(t,x) + \mathcal{O}(h+x)
\end{aligned}$$

and

$$\begin{aligned}
&2\kappa b(0)\partial_x u(t,0) + \kappa\sigma^2(0)\partial_{xx} u(t,0) \\
&= 2\kappa\left(b \cdot \partial_x u - x\partial_x\left(b \cdot \partial_x u\right) + \frac{x^2}{2}\partial_{xx}\left(b \cdot \partial_x u\right)\right) \\
&\quad + \kappa\left(\sigma^2 \cdot \partial_{xx} u - x \cdot \partial_x\left(\sigma^2 \cdot \partial_{xx} u\right)\right) \\
&= 2\kappa b(x)\partial_x u(t,x) + \left(-2x\kappa b(x) + \kappa\sigma^2(x)\right)\partial_{xx} u(t,x) + \mathcal{O}(h+x)
\end{aligned}$$

where we omit function arguments in expansion above. We choose δ small such that $\sigma^2(x) - 2\kappa b(x) \neq 0$ for all $x \in [0, \sigma_2\sqrt{3h}]$ because we assume drift and diffusion terms to be continuous and $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$. This allows us to conclude that $(\sigma^2(x) - 2\kappa b(x))^{-1}$ to be bounded for $x \in [0, \sigma_2\sqrt{3h}]$. This allows us to finish proving the lemma by equating two expressions above and rearranging terms. □

Lemma 4.9. There exists $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_b$, we have,

$$\mathbb{E}_x(\Delta X_k) = \frac{\lambda(x)x^2}{2\sqrt{3h} \cdot \sigma(x)} + \frac{\sigma(x)\lambda(x)\sqrt{3h}}{2} + \lambda(x)b(x)h - x + \mathcal{O}(hx + h^2).$$

where

$$R_1(x) = \frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}} = \mathcal{O}(hx + h^2)$$

Proof. We first calculate $\mathbb{E}_x(X_{k+1})$ where $b(x) < 0$. According to (4.6), given $X_k = x$, X_{k+1} can take four different forms

(1) If $x + \sigma(x)Z_{k+1} \ge 0$ and $x + \sigma(x)Z_{k+1} + b(x)h \ge 0$, then

$$X_{k+1} = x + \sigma(x)Z_{k+1} + b(x)h \quad \text{where} \quad Z_{k+1} \ge -\frac{1}{\sigma(x)}\left(b(x)h + x\right)$$

(2) If $x+\sigma(x)Z_{k+1}\geq 0$ and $x+\sigma(x)Z_{k+1}+b(x)h\leq 0$, then

$$X_{k+1}=-x-\sigma(x)Z_{k+1}-b(x)h \quad \text{where} \quad -\frac{x}{\sigma(x)}\leq Z_{k+1}\leq -\frac{1}{\sigma(x)}\left(b(x)h+x\right)$$

(3) If $x+\sigma(x)Z_{k+1}\leq 0$ and $x+\sigma(x)Z_{k+1}+b(x)h\leq 0$, then

$$X_{k+1}=x+\sigma(x)Z_{k+1}-b(x)h \quad \text{where} \quad \frac{1}{\sigma(x)}\left(b(x)h-x\right)\leq Z_{k+1}\leq -\frac{x}{\sigma(x)}$$

(4) If $x+\sigma(x)Z_{k+1}\leq 0$ and $x+\sigma(x)Z_{k+1}+b(x)h\geq 0$, then

$$X_{k+1}=-x-\sigma(x)Z_{k+1}+b(x)h \quad \text{where} \quad Z_{k+1}\leq \frac{1}{\sigma(x)}\left(b(x)h-x\right)$$

Therefore,

$$\begin{aligned}
\mathbb{E}_x(X_{k+1}) &= \lambda(x)\mathbb{E}_x\left(||x+\sigma(x)Z_{k+1}|+b(x)h|\right)+(1-\lambda(x))\cdot 0\\
&= \int_{-\frac{1}{\sigma(x)}(b(x)h+x)}^{\sqrt{3h}}(x+\sigma(x)y+b(x)h)\cdot\frac{\lambda(x)}{2\sqrt{3h}}\,dy\\
&\quad+\int_{-\frac{x}{\sigma(x)}}^{-\frac{1}{\sigma(x)}(b(x)h+x)}(-x-\sigma(x)y-b(x)h)\cdot\frac{\lambda(x)}{2\sqrt{3h}}\,dy\\
&\quad+\int_{\frac{1}{\sigma(x)}(b(x)h-x)}^{-\frac{x}{\sigma(x)}}(x+\sigma(x)y-b(x)h)\cdot\frac{\lambda(x)}{2\sqrt{3h}}\,dy\\
&\quad+\int_{-\sqrt{3h}}^{\frac{1}{\sigma(x)}(b(x)h-x)}(-x-\sigma(x)y+b(x)h)\cdot\frac{\lambda(x)}{2\sqrt{3h}}\,dy\\
&= \frac{\lambda(x)x^2}{2\sqrt{3h}\cdot\sigma(x)}+\frac{\sigma(x)\lambda(x)\sqrt{3h}}{2}+\lambda(x)b(x)h+\frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}}.
\end{aligned} \tag{A.14}$$

This gives that

$$\mathbb{E}_x(\Delta X_k)=\frac{\lambda(x)x^2}{2\sqrt{3h}\cdot\sigma(x)}+\frac{\sigma(x)\lambda(x)\sqrt{3h}}{2}+\lambda(x)b(x)h-x+\frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}}. \tag{A.15}$$

When the drift term $b(x)$ is positive calculation is simpler (as there is only absolute value on $x+\sigma(x)Z_{k+1}$), we can verify that:

$$\mathbb{E}_x(\Delta X_k)=\frac{\lambda(x)x^2}{2\sqrt{3h}\cdot\sigma(x)}+\frac{\sigma(x)\lambda(x)\sqrt{3h}}{2}+\lambda(x)b(x)h-x. \tag{A.16}$$

In other word, the proof is complete if the $b(x)\geq 0$ and $R_1=0$. Back to the case where $b(x)<0$, the first conditional moment of ΔX_k is

$$\mathbb{E}_x(\Delta X_k)=(\text{A.16})+\frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}},$$

and the remainder

$$R_1(x)=\frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}}=\mathcal{O}(hx+h^2) \tag{A.17}$$

because of Lemma 4.7 and the bounds $b(x)\in[b_1,b_2]$, $\sigma(x)\geq\sigma_1>0$ from Lemma 4.1. □

Lemma 4.10. There exists $\delta > 0$ and $C > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_b$, we have,

$$\mathbb{E}_x(\Delta X_k^2) = -\frac{\lambda(x)}{\sigma(x)\sqrt{3h}} \cdot x^3 + (1+\lambda(x))x^2 - \lambda(x)\sigma(x)\sqrt{3h} \cdot x + \lambda(x)\sigma^2(x) \cdot h + R_2(x)$$

where

$$|R_2(x)| \leq C \cdot (hx + h^2) = \mathcal{O}(hx + h^2).$$

Proof. Again, by Lemma 4.1, after decreasing δ if necessary, every $x \in \Omega_b$ lies in $B_\delta \subset [0, r_0]$. When drift is negative, recycle part of calculation in (A.14) and separate out remainder terms of order greater that 3/2, we have: (when drift is positive, $\frac{b(x)^2 h^2}{\sigma(x)\sqrt{3h}}$ will be removed from $R_2(x)$, but we still reach same conclusion)

$$\begin{aligned}
&\mathbb{E}_x(\Delta X_k^2)\\
&= \lambda(x)\mathbb{E}_x\left((||x+\sigma(x)Z_{k+1}| + b(x)h| - x)^2\right) + (1-\lambda(x))x^2\\
&= \lambda(x)\mathbb{E}_x\left((|x+\sigma(x)Z_{k+1}| + b(x)h)^2 - 2x\,||x+\sigma(x)Z_{k+1}| + b(x)h| + x^2\right) + (1-\lambda(x))x^2\\
&= \lambda(x)\mathbb{E}_x\left(|x+\sigma(x)Z_{k+1}|^2 - 2x\,||x+\sigma(x)Z_{k+1}| + b(x)h| + x^2\right) + (1-\lambda(x))x^2 + R_2'(x)\\
&= \lambda(x)\cdot\left(x^2 + \sigma^2(x)h - 2x\left(\frac{x^2}{2\sigma(x)\sqrt{3h}} + \frac{\sigma(x)\sqrt{3h}}{2} + b(x)h + \frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}\right) + x^2\right)\\
&\quad + (1-\lambda(x))x^2 + R_2'(x)\\
&= \lambda(x)\cdot\left(2x^2 + \sigma^2(x)h - 2x\left(\frac{x^2}{2\sigma(x)\sqrt{3h}} + \frac{\sigma(x)\sqrt{3h}}{2}\right)\right)\\
&\quad + (1-\lambda(x))x^2 - 2x\lambda(x)\left(b(x)h + \frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}\right) + R_2'(x)\\
&= -\frac{\lambda(x)}{\sigma(x)\sqrt{3h}} \cdot x^3 + (1+\lambda(x))x^2 - \lambda(x)\sigma(x)\sqrt{3h} \cdot x + \lambda(x)\sigma^2(x) \cdot h + R_2(x)
\end{aligned}$$

where

$$R_2'(x) = \lambda(x)\mathbb{E}_x\left(b(x)^2h^2 + 2b(x)h\,|x+\sigma(x)Z_{k+1}|\right)$$

and

$$\begin{aligned}
R_2(x) &= -2x\lambda(x)\left(b(x)h + \frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}\right) + R_2'(x)\\
&= 2x\lambda(x)\left(b(x)h + \frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}\right) + \lambda(x)\mathbb{E}_x\left(b(x)^2h^2 + 2b(x)h\,|x+\sigma(x)Z_{k+1}|\right).
\end{aligned}$$

We can bound $R_2(\cdot)$ using the local boundary-layer bounds from Lemma 4.1. Recall that $B_M := |b_1| \vee |b_2|$ and $\sigma(x) \geq \sigma_1 > 0$ on the boundary layer:

$$|R_2(x)| \leq 2B_M hx + \frac{2B_M^2h^2x}{\sigma_1\sqrt{3h}} + B_M^2h^2 + 2B_M h\mathbb{E}_x\left(|x+\sigma(x)Z_{k+1}|\right)\lambda(x).$$

Hence, it suffices to complete the proof by showing that $h\mathbb{E}_x\left(|x+\sigma(x)Z_{k+1}|\right)\lambda(x) = \mathcal{O}(hx+h^2)$. By Lemma 4.7, and using $x \le \sigma_2\sqrt{3h}$ and $|Z_{k+1}| \le \sqrt{3h}$, we have

$$\begin{aligned} h\mathbb{E}_x\left(|x+\sigma(x)Z_{k+1}|\right)|\lambda(x)| &\le h\left(x+\sigma_2\sqrt{3h}\right)|\lambda(x)| \\ &\le Ch^{3/2}\left(\frac{x}{\sqrt{h}}+\sqrt{h}\right) \\ &\le C(hx+h^2). \end{aligned}$$

□

Lemma 4.11. Assume that $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, there exist $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_b$

$$2\left(\mathbb{E}_x(\Delta X_k) - hb(x)\right)\cdot(x+M(x)) + \mathbb{E}_x(\Delta X_k^2) = \sigma^2(x)h + R_3(x) \tag{A.18}$$

where $|R_3(x)| \le C(hx+h^2) = \mathcal{O}(hx+h^2)$ for some constant C independent of x and h. The constant depends only on the boundary-layer bounds $b_1, b_2, \sigma_1, \sigma_2$, κ, and the lower bound of $|\sigma^2(x) - 2\kappa b(x)|$ in a sufficiently small neighborhood of 0.

Proof. Let

$$\mathbb{E}_x(\Delta X_k) = A_h(x) + R_1(x),$$

where $A_h(x)$ denotes the principal term from (A.16). By Lemma 4.7,

$$|R_1(x)| \le C_1(hx+h^2) \tag{A.19}$$

for some constant C_1 independent of h.

Likewise, by Lemma 4.10,

$$\mathbb{E}_x(\Delta X_k^2) = B_h(x) + R_2(x),$$

where

$$B_h(x) = -\frac{\lambda(x)}{\sigma(x)\sqrt{3h}}x^3 + (1+\lambda(x))x^2 - \lambda(x)\sigma(x)\sqrt{3h}\,x + \lambda(x)\sigma^2(x)h,$$

and

$$|R_2(x)| \le C_2(hx+h^2) \tag{A.20}$$

for some constant C_2 independent of h. We now substitute these expansions into the quantity of interest:

$$2\big(\mathbb{E}_x(\Delta X_k) - hb(x)\big)(x+M(x)) + \mathbb{E}_x(\Delta X_k^2),$$

where

$$M(x) = \frac{\kappa\sigma^2(x)}{\sigma^2(x) - 2\kappa b(x)}.$$

This gives

$$\begin{aligned} &2\big(\mathbb{E}_x(\Delta X_k) - hb(x)\big)(x+M(x)) + \mathbb{E}_x(\Delta X_k^2) \\ &\quad= \underbrace{2\big(A_h(x) - hb(x)\big)(x+M(x)) + B_h(x)}_{=:\mathcal{A}_h(x)} + 2R_1(x)(x+M(x)) + R_2(x). \end{aligned} \tag{A.21}$$

A direct rearrangement of $\mathcal{A}_h(x)$ yields

$$\begin{aligned} \mathcal{A}_h(x) = \lambda(x)&\left(\left(\frac{M(x)}{\sigma(x)\sqrt{3h}}+1\right)x^2 + M(x)\sigma(x)\sqrt{3h} + \sigma^2(x)h + 2b(x)M(x)h\right) \\ &- x^2 - 2M(x)x - 2b(x)M(x)h + 2x(\lambda(x)-1)b(x)h. \end{aligned} \tag{A.22}$$

By the definition (4.7) of $\lambda(x)$, the bracketed terms are chosen precisely so that

$$\lambda(x)\left(\left(\frac{M(x)}{\sigma(x)\sqrt{3h}}+1\right)x^2+M(x)\sigma(x)\sqrt{3h}+\sigma^2(x)h+2b(x)M(x)h\right)$$
$$-x^2-2M(x)x-2b(x)M(x)h=\sigma^2(x)h.$$

Hence

$$\mathcal{A}_h(x)=\sigma^2(x)h+2x(\lambda(x)-1)b(x)h. \tag{A.23}$$

Substituting (A.23) into (A.21), we obtain

$$2\big(\mathbb{E}_x(\Delta X_k)-hb(x)\big)(x+M(x))+\mathbb{E}_x(\Delta X_k^2)=\sigma^2(x)h+R_3(x),$$

where

$$R_3(x):=2x(\lambda(x)-1)b(x)h+2R_1(x)(x+M(x))+R_2(x). \tag{A.24}$$

It remains to bound $R_3(x)$. Since $b(0)\neq\sigma^2(0)/(2\kappa)$, continuity implies that, for sufficiently small $\delta>0$, the quantity

$$\sigma^2(x)-2\kappa b(x)$$

is bounded away from 0 on $x\in[0,\sigma_2\sqrt{3h}]$ whenever $h<\delta$. Therefore $M(x)$ is uniformly bounded on this region. Using also the boundary-layer bound $b(x)\in[b_1,b_2]$, the fact that $|\lambda(x)-1|\le 1$, the remainder bounds (A.19)–(A.20), and $x\in[0,\sigma_2\sqrt{3h}]$, we conclude that

$$|R_3(x)|\le C(hx+h^2)$$

for some constant C independent of h. This proves the claim. □

Lemma 4.12. Assume that $b(0)\neq\frac{\sigma^2(0)}{2\kappa}$, there exists $\delta>0$ such that $\forall h<\delta$ and $\forall x\in\Omega_b$,

$$|\mathbb{E}_x(\Delta X_k)|\le C(h+hx+h^2), \tag{A.25}$$

where $C>0$ is independent of x and h and depends only on κ, the local boundary bounds for b,σ, and the nondegeneracy constant coming from $b(0)\neq\sigma^2(0)/(2\kappa)$.

Proof. By Lemma 4.1, after decreasing δ if necessary, all $x\in[0,\sigma_2\sqrt{3h}]$ lie in the fixed boundary neighborhood where the local coefficient bounds apply. Since

$$b(0)\neq\frac{\sigma^2(0)}{2\kappa},$$

continuity implies that, after decreasing δ if necessary, there exists $c_0>0$ such that

$$|\sigma^2(x)-2\kappa b(x)|\ge c_0,\qquad x\in[0,\sigma_2\sqrt{3h}].$$

Hence $M(x)$ is uniformly bounded on this region. Moreover, since

$$M(0)=\frac{\kappa\sigma^2(0)}{\sigma^2(0)-2\kappa b(0)}\neq 0,$$

continuity also implies that there exists $m_0>0$ such that

$$|M(x)|\ge m_0,\qquad x\in[0,\sigma_2\sqrt{3h}]$$

for all sufficiently small h. Decreasing δ once more so that

$$\sigma_2\sqrt{3h} \le \frac{m_0}{2}, \qquad h < \delta,$$

we obtain

$$|x + M(x)| \ge |M(x)| - x \ge \frac{m_0}{2}, \qquad x \in [0, \sigma_2\sqrt{3h}].$$

We derive from remainder estimation in Lemma 4.11 and the crude bound $\mathbb{E}_x(\Delta X_k^2) \le Ch$ that

$$\begin{aligned}
|\mathbb{E}_x(\Delta X_k)| &= \left|\frac{\sigma^2(x)h + R_3(x) - \mathbb{E}_x(\Delta X_k^2)}{2\,(x + M(x))} + hb(x)\right| \\
&\le \frac{\sigma_2^2 h + |R_3(x)| + (\sigma_2\sqrt{3h} + B_M h)^2}{m_0} + B_M h \\
&\le C(h + hx + h^2).
\end{aligned}$$

□

Lemma 4.13. There exists $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_{br}$,

$$\left|\mathbb{E}_x(\Delta X_k^3)\right| \le K_4(hx + h^2), \tag{A.26}$$

where K_4 is a fixed constant dependent on drift, diffusion, and κ.

Proof. By Lemma 4.1, after decreasing δ if necessary, all $x \in [0, \sigma_2\sqrt{3h}]$ lie in the fixed boundary neighborhood where the local coefficient bounds apply. We begin with the case where $x - \sigma(x)\sqrt{3h} \le 0$.

Similar to proof of Lemma 3.8, fix $x \in [0, \sigma_2\sqrt{3h}]$, an increment ΔX_k is supported on $[-x, \sigma_2\sqrt{3h} + B_M h]$. We decompose ΔX_k into its positive and negative parts, so that $|\Delta X_k|^3 = (\Delta X_k^+)^3 + (\Delta X_k^-)^3$ and therefore

$$\mathbb{E}_x(|\Delta X_k|^3) = \mathbb{E}_x[(\Delta X_k^+)^3] + \mathbb{E}_x[(\Delta X_k^-)^3].$$

Positive part. Whenever $\Delta X_k > 0$ the increment comes from the continuous branch Z_k in Algorithm 2. This means the largest positive increment is $\sigma(x)\sqrt{3h} + b(x)h$, hence

$$\mathbb{E}_x[(\Delta X_k^+)^3] \le \lambda(x)\,(\sigma_2\sqrt{3h} + B_M h)^3. \tag{A.27}$$

Recall by Lemma 4.7, for $x \in [0, \sqrt{3h}]$, we have the following bound on $\lambda(x)$,

$$|\lambda(x)| \le \frac{2\sigma_2^2}{\sqrt{3}\sigma_1^3} \cdot \frac{x}{\sqrt{h}} + \frac{(4\sigma_2^4 + 6\kappa B_M \sigma_2^2)}{\sqrt{3}\sigma_1^3\kappa} \cdot \sqrt{h} \le S_1\left(\frac{x}{\sqrt{h}} + \sqrt{h}\right).$$

Combining with (A.27) gives

$$\begin{aligned}
\mathbb{E}_x[(\Delta X_k^+)^3] &\le S_1\left(\frac{x}{\sqrt{h}} + \sqrt{h}\right)(\sigma_2\sqrt{3h} + B_M h)^3 \\
&\le S_1\left(\sqrt{3}\sigma_2 x + B_M\sqrt{h}x + \sqrt{3}\sigma_2 h + B_M h^{3/2}\right)(\sigma_2\sqrt{3h} + B_M h)^2.
\end{aligned}$$

In other word, exist constant $K_4^+ > 0$ that

$$\mathbb{E}_x[(\Delta X_k^+)^3] \le K_4^+(hx + h^2). \tag{A.28}$$

Negative part. Because $\Delta X_k \geq -x$, we have $0 \leq \Delta X_k^- \leq x$, and therefore

$$(\Delta X_k^-)^3 \leq x^3.$$

Since $x \in [0, \sigma_2\sqrt{3h}]$,

$$\mathbb{E}_x[(\Delta X_k^-)^3] \leq x^3 \leq 3\sigma_2^2 hx. \tag{A.29}$$

Combining (A.28) and (A.29) gives

$$\mathbb{E}_x(|\Delta X_k|^3) \ \leq\ K_4^+ h^2 + (K_4^+ + 3\sigma_2^2)hx.$$

Since $|\mathbb{E}_x(\Delta X_k^3)| \leq \mathbb{E}_x(|\Delta X_k|^3)$, the same bound holds for $|\mathbb{E}_x(\Delta X_k^3)|$ with $K_4 = K_4^+ + 3\sigma_2^2$. We have shown the lemma for $x \in \Omega_b$. It remains to consider $x \in \Omega_r$. In this region the scheme performs the reflected Euler–Maruyama update

$$X_{k+1} = |x + \sigma(x)Z_{k+1} + b(x)h|\,.$$

Let $\Delta X_k^{\mathrm{EM}} := \sigma(x)Z_{k+1} + b(x)h$ denote the unreflected Euler–Maruyama increment. By symmetry of Z_{k+1} and the boundary-layer bounds on b and σ,

$$\left|\mathbb{E}_x\left[\left(\Delta X_k^{\mathrm{EM}}\right)^3\right]\right| = \left|3b(x)\sigma^2(x)h^2 + b^3(x)h^3\right| \leq Ch^2.$$

The reflected and unreflected increments differ only on

$$A := \{x + \sigma(x)Z_{k+1} + b(x)h < 0\}\,.$$

Since $x \in \Omega_r$, Lemma 4.1 gives $x = \mathcal{O}(\sqrt{h})$, and the interval of Z_{k+1} values producing reflection has length $\mathcal{O}(h)$. Because Z_{k+1} is uniform on an interval of length $2\sqrt{3h}$, we have

$$\mathbb{P}_x(A) = \mathcal{O}(\sqrt{h}).$$

On A, both $|\Delta X_k|$ and $|\Delta X_k^{\mathrm{EM}}|$ are $\mathcal{O}(\sqrt{h})$, so

$$\left|\mathbb{E}_x\left[\Delta X_k^3 - \left(\Delta X_k^{\mathrm{EM}}\right)^3\right]\right| \leq Ch^{3/2}\mathbb{P}_x(A) = \mathcal{O}(h^2).$$

Hence

$$|\mathbb{E}_x(\Delta X_k^3)| \leq K_4' h^2 \leq K_4'(hx + h^2),$$

and the proof is complete.

□

Theorem 4.14. Given $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, there exists $\delta > 0$ and $C \geq 0$ depending on the relevant bounds of derivatives of u and on the local boundary constants for b, σ such that $\forall h < \delta$ and $\forall x \in \Omega_b$,

$$|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| \leq C(h^2 + hx) = \mathcal{O}(hx + h^2) \tag{A.30}$$

Proof. We choose δ small enough so that all lemmas we have proved in this section hold for $h < \delta$. In particular, by Lemma 4.1, for all $h < \delta$ and all $x \in \Omega_b$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].$$

These constants are local boundary-layer constants and are independent of x and h. Let u be the solution to the backward equation (4.3). We perform a Taylor expansion of u to 3rd order centered at (t_k, X_k). We write

$u_k = u(t_k, X_k), \partial_x u_k = \partial_x u(t_k, X_k)$ (and similarly for $\partial_{xx} u_k, \partial_{xxx} u_k$), and we write Y_k for a point such that $Y_k \in [X_k, X_{k+1}]$ or $Y_k \in [X_{k+1}, X_k]$, and s_k will be a point in interval $s_k \in [kh, (k+1)h]$). Expanding gives

$$\begin{aligned} u_{k+1} - u_k = \partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2} \partial_{xx} u_k \cdot \Delta X_k^2 \\ + \frac{1}{2} \partial_{tt} u(s_k, X_k) \cdot h^2 + \frac{1}{6} \partial_{xxx} u(t_k, Y_k) \cdot \Delta X_k^3. \end{aligned} \tag{A.31}$$

We are interested in $\mathbb{E}_x(u_{k+1} - u_k)$ and know $x \in \Omega_b$. We now bound the expectation of each of the terms in the Taylor-expansion (A.31). We have

$$\begin{aligned} \mathbb{E}_x |\partial_{tt} u(s_k, X_k) \cdot h^2| &\le h^2 \|\partial_{tt} u\|_\infty \\ \mathbb{E}_x |\partial_{xxx} u(t_k, Y_k) \cdot \Delta X_k^3| &\le C(h^2 + hx) \|u^{(3)}\|_\infty, \end{aligned}$$

where the second inequality follows from Lemma 4.13. We also have, using the backward equation (4.3) and Lemma 4.8,

$$\begin{aligned} &\mathbb{E}_x \left[\partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2} \partial_{xx} u_k \cdot \Delta X_k^2 \right] \\ &= \partial_x u(t_k, x) \left(\mathbb{E}_x(\Delta X_k) - b(x) h \right) + \frac{1}{2} \partial_{xx} u(t_k, x) \left(\mathbb{E}_x(\Delta X_k^2) - \sigma^2(x) h \right) \\ &= \left[(x + M(x)) \left(\mathbb{E}_x(\Delta X_k) - b(x) h \right) + \frac{1}{2} \left(\mathbb{E}_x(\Delta X_k^2) - \sigma^2(x) h \right) \right] \partial_{xx} u(t_k, x) \\ &\quad + \left(\mathbb{E}_x(\Delta X_k) - b(x) h \right) Re(x) \\ &= \frac{1}{2} \partial_{xx} u(t_k, x) R_3(x) + \left(\mathbb{E}_x(\Delta X_k) - b(x) h \right) Re(x). \end{aligned} \tag{A.32}$$

The last step uses Lemma 4.11. Therefore, applying the triangle inequality to (A.31), and then using Lemma 4.12 and remainder estimates in Lemma 4.8, 4.11 shows that there exist constants M_1, M_2, independent of x, t, h, such that

$$\begin{aligned} |\mathbb{E}\left(u_{k+1} - u_k | X_k = x \right)| &\le \frac{1}{2} \|\partial_{xx} u\|_\infty \cdot |R_3(x)| + \left(|\mathbb{E}_x(\Delta X_k)| + B_M h \right) \cdot |Re(x)| \\ &\quad + h^2 \|\partial_{tt} u\|_\infty + C(h^2 + hx) \|u^{(3)}\|_\infty \\ &\le M_1 h^2 + M_2 hx \end{aligned}$$

where $B_M := |b_1| \vee |b_2|$. □

Lemma 4.15. There exists $\delta > 0$ and constant C which depends only on the boundary-layer bounds such that $\forall h < \delta$ and $\forall x \in \Omega_r$,

$$|E_x(\Delta X_k) - b(x) h| \le C(hx + h^2) = \mathcal{O}(hx + h^2)$$

Proof. Recall that

$$\Omega_r := \left\{ x \ge \sigma(x)\sqrt{3h} : \; x - \sigma(x)\sqrt{3h} + b(x) h \le 0 \right\}.$$

For $x \in \Omega_r$ the scheme performs a reflected Euler–Maruyama step,

$$X_{k+1} = \left| x + \sigma(x) Z_{k+1} + b(x) h \right|, \qquad Z_{k+1} \sim \mathrm{Unif}[-\sqrt{3h}, \sqrt{3h}].$$

Hence

$$\mathbb{E}_x(X_{k+1}) =$$
$$\frac{1}{2\sqrt{3h}}\int_{-\frac{x+b(x)h}{\sigma(x)}}^{\sqrt{3h}} (x+\sigma(x)z+b(x)h)\ dz + \frac{1}{2\sqrt{3h}}\int_{-\sqrt{3h}}^{-\frac{x+b(x)h}{\sigma(x)}} (-x-\sigma(x)z-b(x)h)\ dz.$$

A direct calculation gives

$$\mathbb{E}_x(X_{k+1}) = \frac{\sigma(x)\sqrt{3h}}{2} + \frac{(x+b(x)h)^2}{2\sigma(x)\sqrt{3h}}.$$

Therefore

$$\begin{aligned}\mathbb{E}_x(\Delta X_k) - b(x)h &= \frac{\sigma(x)\sqrt{3h}}{2} + \frac{(x+b(x)h)^2}{2\sigma(x)\sqrt{3h}} - x - b(x)h \\ &= \frac{\left(x+b(x)h-\sigma(x)\sqrt{3h}\right)^2}{2\sigma(x)\sqrt{3h}}.\end{aligned}$$

In particular,

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| = \frac{\left(x+b(x)h-\sigma(x)\sqrt{3h}\right)^2}{2\sigma(x)\sqrt{3h}}.$$

Let $B_M := |b_1| \vee |b_2|$. By Lemma 4.1, after decreasing δ if needed, we have

$$\sigma(x) \geq \sigma_1 > 0, \qquad |b(x)| \leq B_M$$

for all $x \in \Omega_r$ and all $h < \delta$. Moreover, since $x \in \Omega_r$,

$$x \geq \sigma(x)\sqrt{3h}, \qquad x + b(x)h \leq \sigma(x)\sqrt{3h}.$$

Hence

$$0 \leq \sigma(x)\sqrt{3h} - x - b(x)h \leq -b(x)h \leq B_M h,$$

and therefore

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| \leq \frac{B_M^2 h^2}{2\sigma(x)\sqrt{3h}} \leq \frac{B_M^2}{2\sigma_1\sqrt{3}}\, h^{3/2}.$$

Finally, $x \in \Omega_r$ also implies

$$x \geq \sigma(x)\sqrt{3h} \geq \sigma_1\sqrt{3h},$$

so

$$h^{3/2} \leq \frac{1}{\sigma_1\sqrt{3}}\, hx.$$

Combining the last two bounds yields

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| \leq \frac{B_M^2}{6\sigma_1^2}\, hx \leq C(hx + h^2)$$

for some constant C independent of h and x. This completes the proof. □

Theorem 4.16. (Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in algorithm 2, situation 2)
Given $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, there exists $\delta > 0$ and $C \geq 0$ depending on the relevant derivative bounds of u and on the local boundary constants for b and σ such that $\forall h < \delta$ and $\forall x \in \Omega_r$,

$$|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| \leq C(h^2 + hx) = \mathcal{O}(hx + h^2) \tag{A.33}$$

Proof. The proof is a simplified version of proof of Theorem 4.14. By Lemma 4.1, after decreasing δ if necessary, for all $h < \delta$ and all $x \in \Omega_r$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].$$

Let $B_M := |b_1| \vee |b_2|$. Recall from regularity assumption and Lemma 4.13,

$$\mathbb{E}_x|\partial_{tt}u(s_k, X_k) \cdot h^2| \leq h^2\|\partial_{tt}u\|_\infty$$
$$\mathbb{E}_x|\partial_{xxx}u(t_k, Y_k) \cdot \Delta X_k^3| \leq C(h^2 + hx)\|u^{(3)}\|_\infty,$$

We also have, using the fact that u solves the backward equation (4.3),

$$\mathbb{E}_x\left[\partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2}\partial_{xx}u_k \cdot (\Delta X_k^2)\right]$$
$$= \partial_x u(t_k, x) \cdot \left(\mathbb{E}_x(\Delta X_k) - b(x)h\right) + \frac{1}{2}\partial_{xx}u(t_k, x) \cdot \left(-\sigma^2(x) \cdot h + \mathbb{E}_x(\Delta X_k^2)\right).$$

Apply Lemma 4.15 and apply the lower bound of $\mathbb{E}_x(X_{k+1})$ to the second conditional moment calculation, we have,

$$|E_x(\Delta X_k) - b(x)h| \leq C_2(hx + h^2)$$
$$|E_x(\Delta X_k^2) - \sigma^2(x)h| \leq C_3(hx + h^2)$$

Therefore, applying the triangle inequality to the Taylor expansion (A.31), and then using the first- and second-moment estimates above and Lemma 4.13, there exist constants M_1, M_2, independent of x, t, h, such that

$$\begin{aligned}|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| &\leq \|\partial_x u\|_\infty \cdot C_2(hx + h^2) + \frac{1}{2}\|\partial_{xx}u\|_\infty \cdot C_3(hx + h^2)\\ &\quad + h^2\|\partial_{tt}u\|_\infty + C_1(h^2 + hx)\|u^{(3)}\|_\infty\\ &\leq M_1 h^2 + M_2 hx.\end{aligned}$$

□

Lemma 4.17. When $b(0) = \frac{\sigma^2(0)}{2\kappa}$, there exist $\delta > 0$ and and constant C which depends only on the boundary-layer bounds such that $\forall h < \delta$ and $\forall x \in \Omega_b$,

$$|\mathbb{E}_x\left(\Delta X_k\right) - hb(x)| \leq C(hx + h^2) = \mathcal{O}(hx + h^2).$$

Proof. As before,we know there exists $\delta > 0$ such that for all $h < \delta$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2], \qquad \lambda(x) \in [0, 1], \qquad \forall x \in \Omega_b.$$

It suffices to complete the proof if we show the inequality holds for all $x \in [0, \sigma_2\sqrt{3h}]$. Since $\sigma(0) > 0$, the assumption $b(0) = \sigma^2(0)/(2\kappa)$ implies $b(0) > 0$. By continuity, after further decreasing δ if necessary, $b(x) \geq 0$ for any $x \in \Omega_b$ and $\Omega_r = \emptyset$. We can check that when $b(x) = \frac{\sigma^2(x)}{2\kappa}$,

$$\mathbb{E}_x\left(\Delta X_k\right) = hb(x),$$

we shall expect that = is replaced by $\approx$ within the boundary layer. We use Taylor expansion to make the argument more accurate, $\mathbb{E}_x(\Delta X_k)$ is a good approximation to $hb(x)$. We know that, denote $\sigma_0 := \sigma(0)$ and $b_0 := b(0)$,

$$\sigma^2(x) = (\sigma_0 + x \cdot \partial_x \sigma(z_x))^2 \,,$$
$$b(x) = b_0 + x \cdot \partial_x b(y_x) \,.$$

Therefore,

$$\sigma^2(x) - 2\kappa b(x) = \sigma_0^2 - 2\kappa b_0 + \hat{R}(x) = \hat{R}(x) \tag{A.34}$$

where

$$|\hat{R}(x)| = \left| (2\sigma_0 \cdot \partial_x \sigma(z_x) - 2\kappa \cdot \partial_x b(y_x))\, x + (\partial_x \sigma(z_x))^2 \cdot x^2 \right| \leq \hat{K}(x + x^2).$$

where $\hat{K}$ depends only on the local first-derivative bounds of b and σ on the boundary neighborhood, and is independent of x and h. We substitute (A.34) into (4.7),

$$\begin{aligned}
\lambda(x) &= \frac{\hat{R}(x)x^2 + 2\kappa\sigma^2(x)x + \sigma^2(x)h \cdot \left(2\kappa b(x) + \hat{R}(x)\right)}{\frac{\kappa\sigma(x)}{\sqrt{3h}} \cdot x^2 + \hat{R}(x) \cdot x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^2(x)h \cdot \left(2\kappa b(x) + \hat{R}(x)\right)} \\
&= \frac{\kappa\sigma^2(x)\,(2x + 2b(x)h + R(x))}{\kappa\sigma^2(x)\left(\frac{x^2}{\sigma(x)\sqrt{3h}} + \sigma(x)\sqrt{3h} + 2b(x)h + R(x)\right)} \\
&= \frac{2x + 2b(x)h + R(x)}{\frac{x^2}{\sigma(x)\sqrt{3h}} + \sigma(x)\sqrt{3h} + 2b(x)h + R(x)}
\end{aligned}$$

where

$$R(x) = \frac{\sigma^2(x)h + x^2}{\kappa\sigma^2(x)} \cdot \hat{R}(x) \,.$$

Recall $x \leq \sigma_2\sqrt{3h}$,

$$\frac{\sigma^2(x)h + x^2}{\kappa\sigma^2(x)} \leq \frac{\sigma_2^2 h + 3\sigma_2^2 h}{k\sigma_1^2} = Ch.$$

Hence,

$$|R(x)| \leq Ch|\hat{R}(x)| \leq Ch(x + x^2) \leq C(hx + h^2).$$

Since the drift is positive, substitute $\lambda(x)$ back into (A.16), we have

$$\begin{aligned}
\mathbb{E}_x(\Delta X_k) &= \lambda(x) \cdot \left(\frac{x^2}{2\sigma(x)\sqrt{3h}} + \frac{1}{2}\sigma(x)\sqrt{3h} + b(x)h + \frac{1}{2}R(x) - \frac{1}{2}R(x)\right) - x \\
&= x + b(x)h + \frac{1}{2}\left(1 - \lambda(x)\right)R(x) - x \\
&= b(x)h + \frac{1}{2}\left(1 - \lambda(x)\right)R(x)
\end{aligned}$$

Hence,

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| = \left|\frac{1}{2}\left(1 - \lambda(x)\right)R(x)\right| \leq C(hx + h^2)$$

because we know $\lambda(x) \in [0, 1]$, $x \in [0, \sigma_2\sqrt{3h}]$, $|\hat{R}(x)| \leq \hat{K}(x + x^2)$, and the coefficient $\frac{\sigma^2(x)h + x^2}{\kappa\sigma^2(x)}$ we scale $\hat{R}(\cdot)$ to $R(\cdot)$ is of order h using the boundary-layer bounds $\sigma_1 \leq \sigma(x) \leq \sigma_2$ from Lemma 4.1. $\square$

Theorem 4.18. (Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in algorithm 2, situation 3)
If $b(0)=\frac{\sigma^2(0)}{2\kappa}$, there exist $\delta>0$ and $C\geq 0$, independent of x and h, depending only on the relevant bounds of derivatives of u and on the local boundary constants for b,σ such that $\forall h<\delta$ and for all $x\in\Omega_{br}$,

$$|\mathbb{E}\left(u_{k+1}-u_k|X_k=x\right)|\leq C(h^2+hx)=\mathcal{O}(hx+h^2). \tag{A.35}$$

Proof. The proof follows the same Taylor-expansion argument as in Theorem 4.14. Fix x such that $x-\sigma(x)\sqrt{3h}<0$. By Lemma 4.1, after decreasing δ if necessary, for all $h<\delta$ and all x satisfying $x-\sigma(x)\sqrt{3h}<0$,

$$b(x)\in[b_1,b_2],\qquad \sigma(x)\in[\sigma_1,\sigma_2].$$

These constants are independent of x and h. By Taylor expansion of u centered at (t_k,x), with Y_k between x and X_{k+1} and $s_k\in[t_k,t_{k+1}]$, we have

$$\begin{aligned}u_{k+1}-u_k&=\partial_t u_k\,h+\partial_x u_k\,\Delta X_k+\frac{1}{2}\partial_{xx}u_k\,\Delta X_k^2\\&\quad+\frac{1}{2}\partial_{tt}u(s_k,x)\,h^2+\frac{1}{6}\partial_{xxx}u(t_k,Y_k)\,\Delta X_k^3.\end{aligned}$$

Taking conditional expectation given $X_k=x$ and using the backward equation (4.3), we obtain

$$\begin{aligned}&|\mathbb{E}\left(u_{k+1}-u_k\mid X_k=x\right)|\\&\leq|\partial_x u(t_k,x)|\cdot|\mathbb{E}_x(\Delta X_k)-b(x)h|+\frac{1}{2}\left|\partial_{xx}u(t_k,x)\right|\cdot\left|\mathbb{E}_x(\Delta X_k^2)-\sigma^2(x)h\right|\\&\quad+\frac{h^2}{2}\|\partial_{tt}u\|_\infty+\frac{1}{6}\|u^{(3)}\|_\infty\left|\mathbb{E}_x(\Delta X_k^3)\right|.\end{aligned}\tag{A.36}$$

Because $b(0)=\frac{\sigma^2(0)}{2\kappa}$, the sticky boundary condition implies

$$\partial_{xx}u(t,0)=0.$$

Hence, by Taylor expansion in space,

$$\partial_{xx}u(t_k,x)=x\,u^{(3)}(t_k,y_x)\,,\qquad\text{for some } y_x\in[0,x].$$

Therefore,

$$|\partial_{xx}u(t_k,x)|\leq\|u^{(3)}\|_\infty\,x.$$

Next, by Lemma 4.17,

$$|\mathbb{E}_x(\Delta X_k)-b(x)h|\leq C_1(hx+h^2).$$

Moreover, since $x-\sigma(x)\sqrt{3h}<0$, the increment is supported in an interval of size $O(\sqrt{h})$, so

$$\mathbb{E}_x(\Delta X_k^2)\leq C_2h$$

for some constant $C_2>0$ independent of x and h because on the boundary layer the update either jumps to 0 or takes a reflected Euler–Maruyama step. Finally, by the third-moment bound already established for the boundary layer,

$$\left|\mathbb{E}_x(\Delta X_k^3)\right|\leq C_3(hx+h^2).$$

Substituting these estimates into the inequality (A.36) yields

$$\begin{aligned}|\mathbb{E}\left(u_{k+1}-u_k \mid X_k=x\right)| &\le \|\partial_x u\|_\infty \cdot C_1(hx+h^2)+\frac{1}{2}\|u^{(3)}\|_\infty x\cdot\left(\sigma^2(x)h+C_2h\right)\\ &\quad+\frac{h^2}{2}\|\partial_{tt}u\|_\infty+\frac{1}{6}\|u^{(3)}\|_\infty C_3(hx+h^2)\\ &\le M_1h^2+M_2hx,\end{aligned}$$

for suitable constants $M_1, M_2 \ge 0$ independent of x and h, after possibly decreasing δ so that all preceding lemmas hold simultaneously. □

Proof of Theorem 4.3. The estimate on Ω_{br} follows from the three boundary-layer results already established. More precisely, when $b(0) \ne \frac{\sigma^2(0)}{2\kappa}$, Theorems 4.14 and 4.16 cover Ω_b and Ω_r, respectively. When $b(0) = \frac{\sigma^2(0)}{2\kappa}$, we have $b(0) > 0$, so by continuity, after possibly decreasing δ, one has $b(x) > 0$ throughout the boundary layer; hence $\Omega_r = \varnothing$, and Theorem 4.18 yields the estimate on all of $\Omega_{br} = \Omega_b$. Therefore, after decreasing $\delta > 0$ finitely many times and enlarging the constant if needed, we may assume that

$$|\mathbb{E}\left(u_{k+1}-u_k \mid X_k=x\right)| \le M_b(h^2+hx), \qquad \forall x\in\Omega_{br},\ \forall h<\delta.$$

We now check $x \in \Omega_e$. Recall that, outside the boundary layer, the simulation scheme makes a standard Euler–Maruyama jump. Since

$$x-\sigma(x)\sqrt{3h}+b(x)h>0,$$

the absolute value in (4.5) is inactive, and therefore

$$\Delta X_k = \sigma(x)Z_{k+1}+b(x)h.$$

Hence

$$\mathbb{E}_x(\Delta X_k)=b(x)h, \qquad \mathbb{E}_x(\Delta X_k^2)=\sigma^2(x)h+b^2(x)h^2,$$
$$\mathbb{E}_x(\Delta X_k^3)=3b(x)\sigma^2(x)h^2+b^3(x)h^3,$$

and

$$\begin{aligned}\mathbb{E}_x(\Delta X_k^4)&=\mathbb{E}_x\left(\sigma(x)Z_{k+1}+b(x)h\right)^4\\ &=\frac{9}{5}\sigma^4(x)h^2+6b^2(x)\sigma^2(x)h^3+b^4(x)h^4.\end{aligned}\tag{A.37}$$

The Taylor expansion upto fourth order is

$$\begin{aligned}u_{k+1}-u_k&=\partial_t u_k\cdot h+\partial_x u_k\cdot\Delta X_k+\frac{1}{2}\partial_{xx}u_k\cdot\Delta X_k^2\\ &\quad+\frac{1}{2}\partial_{tt}u(s_k,X_k)\cdot h^2+\frac{1}{6}\partial_{xxx}u_k\cdot\Delta X_k^3+\frac{1}{24}\partial_{xxxx}u(t_k,Y_k)\cdot\Delta X_k^4.\end{aligned}\tag{A.38}$$

Using the fact that u solves the backward equation (4.3), we know

$$\begin{aligned}&\mathbb{E}_x\left[\partial_t u_k\cdot h+\partial_x u_k\cdot\Delta X_k+\frac{1}{2}\partial_{xx}u_k\cdot(\Delta X_k^2)\right]\\ &=-b(x)\partial_x u(t_k,x)\cdot h-\frac{\sigma^2(x)}{2}\partial_{xx}u(t_k,x)\cdot h+\partial_x u_k\cdot b(x)h+\frac{1}{2}\partial_{xx}u(t_k,x)\cdot\left(\sigma^2(x)h+b^2(x)h^2\right)\\ &=\frac{1}{2}b^2(x)\partial_{xx}u(t_k,x)\,h^2.\end{aligned}$$

Taking conditional expectation in (A.38), and using the previous calculation, gives

$$
\begin{aligned}
|\mathbb{E}\left(u_{k+1}-u_k \mid X_k=x\right)| \leq\ & \frac{1}{2}|b(x)|^2\|\partial_{xx}u\|_\infty h^2+\frac{h^2}{2}\|\partial_{tt}u\|_\infty \\
&+\frac{1}{6}\|u^{(3)}\|_\infty\left|3b(x)\sigma^2(x)h^2+b^3(x)h^3\right| \\
&+\frac{1}{24}\|u^{(4)}\|_\infty\left|\frac{9}{5}\sigma^4(x)h^2+6b^2(x)\sigma^2(x)h^3+b^4(x)h^4\right|.
\end{aligned}
$$

By Assumption 4, there exists $L>0$ such that

$$
|b(x)|+|\sigma(x)| \leq L(1+x), \qquad x \geq 0.
$$

Hence, using $h<1$ and $(1+x)^4 \leq C(1+x^4)$ for $x \geq 0$, we have

$$
|b(x)|^2+|b(x)|\,\sigma^2(x)+|b(x)|^3 h \leq C(1+x^3) \leq C(1+x^4),
$$

and

$$
\sigma^4(x)+b^2(x)\sigma^2(x)h+b^4(x)h^2 \leq C(1+x^4).
$$

Substituting these bounds and (A.37) into the previous estimate yields

$$
|\mathbb{E}\left(u_{k+1}-u_k \mid X_k=x\right)| \leq M_e h^2(1+x^4), \qquad \forall x \in \Omega_e,\ \forall h<\delta,
$$

for some constant M_e independent of x and h. □

Lemma 4.19. Denote $\tilde{\lambda}(\cdot)$ as the jump probability we propose in algorithm 1 and $\lambda(\cdot)$ as the one we propose in algorithm 2, if for the diffusion $\sigma(0)=1$ for a sticky diffusion, then

$$
\left|\lambda(x)-\tilde{\lambda}(x)\right| \leq K_{b,\sigma}\sqrt{h}, \quad \forall x \in [0,\sqrt{3h}]
$$

where $K_{b,\sigma}>0$ is a constant depending on the drift term $b(\cdot)$, diffusion $\sigma(\cdot)$, and sticky parameter κ.

Proof. Recall that

$$
\lambda(x)=\frac{N_\sigma(x)}{D_\sigma(x)},
$$

where

$$
N_\sigma(x) := (\sigma^2(x)-2\kappa b(x))x^2+2\kappa\sigma^2(x)x+\sigma^4(x)h
$$

and

$$
D_\sigma(x) := \left(\frac{\kappa\sigma(x)}{\sqrt{3h}}+\sigma^2(x)-2\kappa b(x)\right)x^2+\sigma^3(x)\kappa\sqrt{3h}+\sigma^4(x)h.
$$

Similarly,

$$
\tilde{\lambda}(x)=\frac{N_0(x)}{D_0(x)},
$$

where

$$
N_0(x) := x^2+2\kappa x+h, \qquad D_0(x) := \left(\frac{\kappa}{\sqrt{3h}}+1\right)x^2+h+\kappa\sqrt{3h}.
$$

Using the same compactness argument as in Lemma 4.1, and using $\sigma(0)=1$, there exist $\delta_1>0$ and constants $b_1, b_2, \sigma_1, \sigma_2$, with $\sigma_1>0$, such that for every $h<\delta_1$ and every $x \in [0,\sqrt{3h}]$,

$$
b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].
$$

Set

$$B_M := |b_1| \vee |b_2|.$$

Moreover, since $\sigma(0) = 1$ and σ is continuously differentiable near 0, after decreasing δ_1 if necessary, there exists a constant $C_\sigma > 0$, independent of x and h, such that for $n = 1, 2, 3, 4$,

$$|\sigma^n(x) - 1| \le C_\sigma x \le C_\sigma \sqrt{3h}, \qquad \forall x \in [0, \sqrt{3h}], \quad h < \delta_1. \tag{A.39}$$

We also decrease δ_1 so that $\delta_1 \le 1$ and, for every $h < \delta_1$,

$$\frac{\kappa\sigma_1}{\sqrt{3h}} > \sigma_2^2 + 2\kappa B_M.$$

Then, for every $x \in [0, \sqrt{3h}]$,

$$D_\sigma(x) \ge \sigma^3(x)\kappa\sqrt{3h} \ge \kappa\sigma_1^3\sqrt{3h}. \tag{A.40}$$

Now define

$$\hat{\lambda}(x) := \frac{N_0(x)}{D_\sigma(x)}.$$

We first compare $\lambda(x)$ and $\hat{\lambda}(x)$. Since

$$N_\sigma(x) - N_0(x) = (\sigma^2(x) - 1)x^2 - 2\kappa b(x)x^2 + 2\kappa(\sigma^2(x) - 1)x + (\sigma^4(x) - 1)h,$$

using (A.39) and $x \le \sqrt{3h}$ gives

$$|N_\sigma(x) - N_0(x)| \le C_N h \tag{A.41}$$

for some constant $C_N > 0$ independent of x and h. Hence, by (A.40),

$$|\lambda(x) - \hat{\lambda}(x)| = \frac{|N_\sigma(x) - N_0(x)|}{D_\sigma(x)} \le \frac{C_N h}{\kappa\sigma_1^3\sqrt{3h}} \le C_N'\sqrt{h},$$

where $C_N' > 0$ is independent of x and h.

Next, we compare $\hat{\lambda}(x)$ and $\tilde{\lambda}(x)$. We have

$$D_\sigma(x) - D_0(x) = \left(\frac{\kappa(\sigma(x) - 1)}{\sqrt{3h}} + \sigma^2(x) - 1 - 2\kappa b(x)\right)x^2 + \kappa(\sigma^3(x) - 1)\sqrt{3h} + (\sigma^4(x) - 1)h.$$

Using (A.39) again, together with $x \le \sqrt{3h}$ and $|b(x)| \le B_M$, we obtain

$$|D_\sigma(x) - D_0(x)| \le C_D h \tag{A.42}$$

for some constant $C_D > 0$ independent of x and h. Moreover,

$$\hat{\lambda}(x) - \tilde{\lambda}(x) = \frac{D_0(x) - D_\sigma(x)}{D_\sigma(x)} \cdot \frac{N_0(x)}{D_0(x)}.$$

Using Lemma 3.1, which gives $0 \le \tilde{\lambda}(x) \le 1$ on $[0, \sqrt{3h}]$, together with (A.40) and (A.42), we obtain

$$|\hat{\lambda}(x) - \tilde{\lambda}(x)| \le \frac{|D_\sigma(x) - D_0(x)|}{D_\sigma(x)}|\tilde{\lambda}(x)| \le \frac{C_D h}{\kappa\sigma_1^3\sqrt{3h}} \le C_D'\sqrt{h},$$

where $C_D' > 0$ is independent of x and h.

Combining the two estimates, for every $h < \delta_1$ and every $x \in [0, \sqrt{3h}]$,

$$|\lambda(x) - \tilde{\lambda}(x)| \leq |\lambda(x) - \hat{\lambda}(x)| + |\hat{\lambda}(x) - \tilde{\lambda}(x)| \leq K_{b,\sigma}\sqrt{h}.$$

Here $K_{b,\sigma} := C'_N + C'_D$ depends only on κ, the local bounds $b_1, b_2, \sigma_1, \sigma_2$, and the local derivative bound for σ, but is independent of x and h. This completes the proof with $\delta = \delta_1$. ☐

Lemma 4.4. There exists $\delta > 0$ and constant C such that for all $h < \delta$,

$$\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \cdot \mathbf{1}_{\Omega_{br}}(X_k)\right) < C$$

Proof. Throughout this proof we use the normalization $\sigma(0) = 1$ made before the statement of the lemma. By the proof of Lemma 4.1, there exist $\delta_B > 0$ and a constant $K_B > 0$, depending only on local bounds for b and σ near 0, such that for every $h < \delta_B$,

$$\Omega_{br} \subset B_h := [0, \sqrt{3h} + K_B h]. \tag{A.43}$$

Hence it is enough to prove a uniform bound with $\mathbf{1}_{B_h}$ in place of $\mathbf{1}_{\Omega_{br}}$. The rest of the proof is a discrete supersolution argument. We will construct a bounded barrier function $\varphi_h = e^{w_h}$ whose one-step drift is negative by an amount comparable to x on the enlarged boundary layer B_h. More precisely, the key estimate is

$$P\varphi_h(x) \leq \varphi_h(x)(1 - f_h(x) + C_1 h), \qquad f_h(x) \geq x\mathbf{1}_{B_h}(x).$$

Once this is proved, the time factor $e^{K_T(T-t_i)}$ will absorb the harmless $C_1 h$ error, giving a supersolution for the discrete occupation equation. A backward comparison then bounds the expected occupation sum.

We now modify the test function used in Lemma 3.5 so that it is unchanged near the boundary but becomes constant away from the boundary. This avoids requiring b and σ to be globally bounded. Fix $s > 16$ and $M > 0$. Choose a fixed number $R > 1$, independent of h. Later we decrease δ_B if necessary so that

$$B_h \subset [0, R/4].$$

Let $\chi \in C^2(\mathbb{R})$ satisfy

$$0 \leq \chi \leq 1, \qquad \chi(r) = 1 \text{ for } r \leq 0, \qquad \chi(r) = 0 \text{ for } r \geq 1.$$

Define

$$\eta_R(x) := \int_0^x \chi\left(\frac{y - R}{R}\right) dy.$$

Then $\eta_R(x) = x$ for $0 \leq x \leq R$, while $\eta_R(x) = A_R$ for $x \geq 2R$, where

$$A_R := \int_0^{2R} \chi\left(\frac{y - R}{R}\right) dy.$$

Moreover, $0 \leq \eta'_R(x) \leq 1$ and $|\eta''_R(x)| \leq C_R$ for a constant C_R depending only on R and the cutoff χ.

Define

$$w_h(x) := \begin{cases} M, & 0 \leq x \leq \sqrt{3h}/2, \\ M - s\left(\eta_R(x) - \frac{\sqrt{3h}}{2}\right), & x > \sqrt{3h}/2. \end{cases} \tag{A.44}$$

Then w_h agrees with the function used in Lemma 3.5 on $[0,R]$:

$$w_h(x) = \begin{cases} M, & 0 \le x \le \sqrt{3h}/2, \\ M - s\left(x - \frac{\sqrt{3h}}{2}\right), & \sqrt{3h}/2 < x \le R. \end{cases}$$

Also, w_h is constant on $[2R,\infty)$. Let $\varphi_h(x) := e^{w_h(x)}$. Since $w_h(x) \le M$ for all $x \ge 0$, $\varphi_h(x) \le e^M$ uniformly in x and h. We first prove that there exist constants $C_1 > 0$ and $\delta_1 > 0$, independent of x and h, such that for all $h < \delta_1$,

$$P\varphi_h(x) \le \varphi_h(x)\left(1 - f_h(x) + C_1 h\right), \qquad \forall x \ge 0, \tag{A.45}$$

where f_h satisfies

$$f_h(x) \ge x\mathbf{1}_{B_h}(x). \tag{A.46}$$

We introduce an auxiliary transition operator P_0. The role of P_0 is to separate the genuinely new diffusion effects from the SBM calculation. On B_h, the normalization $\sigma(0) = 1$ implies $\sigma(x) - 1 = O(\sqrt{h})$ and the scheme makes uniformly distributed jump of size $O(\sqrt{h})$, so the continuous update in the diffusion scheme differs from the SBM reflected update by only $O(h)$. The remaining complication is that P and P_0 correspond to different boundary layer . Given $Z \sim \mathcal{U}[-\sqrt{3h}, \sqrt{3h}]$,

$$P_0\varphi_h(x) = \begin{cases} \lambda(x)\mathbb{E}\left[\varphi_h(|x+Z|)\right] + (1-\lambda(x))\varphi_h(0), & 0 \le x \le \sqrt{3h}, \\ \mathbb{E}\left[\varphi_h(|x+Z|)\right], & x > \sqrt{3h}. \end{cases}$$

Thus P_0 is the same as the SBM transition operator, except that it uses the diffusion jump probability $\lambda(x)$ instead of $\tilde{\lambda}(x)$. We claim that for $x \in B_h$,

$$P\varphi_h(x) \le P_0\varphi_h(x) + C_2 h\varphi_h(x), \tag{A.47}$$

where C_2 is independent of x and h. We prove this by coupling the true diffusion update and the auxiliary update using the same random variable $Z \sim \mathcal{U}[-\sqrt{3h}, \sqrt{3h}]$.

First, after decreasing δ_1 if necessary, $B_h \subset [0, R/4]$. Since $\sigma(0) = 1$ and σ is locally Lipschitz near 0, there exists a constant $C_\sigma > 0$, independent of x and h, such that

$$|\sigma(x) - 1| \le C_\sigma x \le C_\sigma(\sqrt{3h} + K_B h) \le C\sqrt{h}, \qquad x \in B_h.$$

Similarly, b is locally bounded on a fixed compact neighborhood of 0, so

$$|b(x)| \le C, \qquad x \in B_h.$$

Thus, for all possible values of Z and all $x \in B_h$,

$$|x + Z| \le x + \sqrt{3h} \le C\sqrt{h},$$

and

$$\left||x + \sigma(x)Z| + b(x)h\right| \le x + |\sigma(x)Z| + |b(x)|h \le C\sqrt{h}.$$

After decreasing δ_1 once more so that $C\sqrt{h} < R$, all one-step positions appearing in the comparison below lie in $[0,R]$. We also record a bound on the Lipschitz constant of φ_h. Since $0 \le \eta_R' \le 1$, the function η_R is 1-Lipschitz. For h small enough that $\sqrt{3h}/2 < R$, we can write

$$w_h(x) = M - s\left(\eta_R(x) - \frac{\sqrt{3h}}{2}\right)_+,$$

Hence w_h is s-Lipschitz, uniformly in h. Since $w_h \le M$,

$$|\varphi_h(y) - \varphi_h(z)| = |e^{w_h(y)} - e^{w_h(z)}| \le e^M |w_h(y) - w_h(z)| \le s e^M |y - z|. \tag{A.48}$$

Therefore φ_h is Lipschitz with constant $L_\varphi := se^M$, independent of x and h. Finally, on B_h we also have a uniform lower bound on $\varphi_h(x)$. Indeed, after decreasing δ_1 if necessary,

$$\varphi_h(x) = \exp(w_h(x)) \ge \exp\left(M - s\left(\frac{\sqrt{3h}}{2} + K_B h\right)\right) \ge e^{M/2}, \qquad x \in B_h. \tag{A.49}$$

Consequently, any estimate of the form Ch on B_h may be rewritten as $Ch\varphi_h(x)$ after increasing the constant C. We now compare the continuous branches. Let

$$Y_0 := |x + Z|$$

denote the continuous update in the auxiliary operator P_0. For the true diffusion scheme, define the continuous update

$$Y_1 := ||x + \sigma(x)Z| + b(x)h|$$

Using $||a| - |b|| \le |a - b|$ gives

$$\begin{aligned}
|Y_1 - Y_0| &= |||x + \sigma(x)Z| + b(x)h| - |x + Z|| \\
&\le ||x + \sigma(x)Z| + b(x)h - |x + Z|| \\
&\le ||x + \sigma(x)Z| - |x + Z|| + |b(x)|h \\
&\le |(\sigma(x) - 1)Z| + |b(x)|h \\
&\le Ch.
\end{aligned} \tag{A.50}$$

Therefore,

$$|\varphi_h(Y_1) - \varphi_h(Y_0)| \le L_\varphi |Y_1 - Y_0| \le Ch \le Ch\varphi_h(x). \tag{A.51}$$

It remains to control the possible mismatch between the two boundary thresholds $\sigma(x)\sqrt{3h}$ and $\sqrt{3h}$. A mismatch occurs only when one of the inequalities

$$x \le \sigma(x)\sqrt{3h}, \qquad x \le \sqrt{3h}$$

holds and the other fails. Equivalently, x lies between $\sqrt{3h}$ and $\sigma(x)\sqrt{3h}$. Hence

$$|x - \sigma(x)\sqrt{3h}| \le |\sigma(x)\sqrt{3h} - \sqrt{3h}| = |\sigma(x) - 1|\sqrt{3h} \le Ch. \tag{A.52}$$

In this mismatch region, we also need that $1 - \lambda(x)$ is of order h. Using the identity

$$1 - \lambda(x) = \frac{\frac{\kappa\sigma(x)}{\sqrt{3h}}\left(x - \sigma(x)\sqrt{3h}\right)^2}{\left(\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x)\right)x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^4(x)h}, \tag{A.53}$$

and using the same denominator lower-bound argument as in Lemma 4.19, after decreasing δ_1 if necessary, there exists $c > 0$ such that

$$\left(\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x)\right)x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^4(x)h \ge c\sqrt{h}, \qquad x \in B_h.$$

Combining this lower bound with (A.52) and (A.53), we obtain

$$1-\lambda(x) \le \frac{C\frac{1}{\sqrt{h}}\cdot h^2}{c\sqrt{h}} \le Ch. \tag{A.54}$$

We now consider the possible threshold configurations. The following four cases only account for which operator regards x as being in the sticky boundary layer. If both operators make the same decision, the comparison follows from the $O(h)$ continuous-branch estimate. If their decisions differ, then x lies in the thin mismatch region, where $1-\lambda(x)=O(h)$, so the extra sticky-jump contribution is also only $O(h)$.

(1) Suppose $x \in \Omega_b$ and $x \le \sqrt{3h}$. Then

$$P\varphi_h(x) = \lambda(x)\mathbb{E}\left[\varphi_h(Y_1)\right] + (1-\lambda(x))\varphi_h(0),$$

and

$$P_0\varphi_h(x) = \lambda(x)\mathbb{E}\left[\varphi_h(Y_0)\right] + (1-\lambda(x))\varphi_h(0). \tag{A.55}$$

Thus, by (A.51)

$$P\varphi_h(x) - P_0\varphi_h(x) = \lambda(x)\mathbb{E}\left[\varphi_h(Y_1)-\varphi_h(Y_0)\right] \le Ch\varphi_h(x).$$

(2) Suppose that $x \in \Omega_b$ and $x > \sqrt{3h}$, then P_0 uses only the continuous branch,

$$\begin{aligned} P\varphi_h(x) - P_0\varphi_h(x) &= \lambda(x)\mathbb{E}\left[\varphi_h(Y_1)-\varphi_h(Y_0)\right] + (1-\lambda(x))\left(\varphi_h(0) - \mathbb{E}\left[\varphi_h(Y_0)\right]\right) \\ &\le Ch\varphi_h(x) + (1-\lambda(x))\cdot C\varphi_h(x) \\ &\le Ch\varphi_h(x). \end{aligned}$$

where we use (A.51), (A.54) and the fact that

$$|\varphi_h(0)-\varphi_h(Y_0)| \le 2e^M \le 2e^{M/2}\varphi_h(x) \le C\varphi_h(x), \qquad x \in B_h,$$

which follows from the upper bound $\varphi_h \le e^M$ and the lower bound (A.49).

(3) Suppose that $x \notin \Omega_b$ and $x \le \sqrt{3h}$, then the true diffusion scheme uses only the continuous branch, while P_0 is in its sticky layer. Hence

$$\begin{aligned} P\varphi_h(x) - P_0\varphi_h(x) &= \mathbb{E}\left[\varphi_h(Y_1)-\varphi_h(Y_0)\right] + (1-\lambda(x))\left(\mathbb{E}\left[\varphi_h(Y_0)\right] - \varphi_h(0)\right) \\ &\le Ch\varphi_h(x) + (1-\lambda(x))\cdot C\varphi_h(x) \\ &\le Ch\varphi_h(x). \end{aligned}$$

(4) Suppose that $x \notin \Omega_b$ and $x \ge \sqrt{3h}$, then both operators use only their continuous branch, and therefore

$$P\varphi_h(x) - P_0\varphi_h(x) = \mathbb{E}\left[\varphi_h(Y_1)-\varphi_h(Y_0)\right] \le Ch\varphi_h(x).$$

Taking C_2 large enough proves (A.47).

We now estimate $P_0\varphi_h$ on $[0,\sqrt{3h}]$. On this interval w_h agrees with the test function in Lemma 3.5. The only difference between P_0 and the SBM transition operator is the replacement of $\tilde{\lambda}(x)$ by $\lambda(x)$. Let $\tilde{P}_0$ denote the same auxiliary operator as P_0, but with λ replaced by $\tilde{\lambda}$. We wish to show $P_0\varphi_h(x) \approx \tilde{P}_0\varphi_h(x)$ so that we can reuse calculation done in Lemma 3.5. For $x \in [0,\sqrt{3h}]$, together with (A.55),

$$P_0\varphi_h(x) - \tilde{P}_0\varphi_h(x) = \left(\lambda(x) - \tilde{\lambda}(x)\right)\left(\mathbb{E}\left[\varphi_h(Y_0)\right] - \varphi_h(0)\right),$$

where $Y_0 = |x + Z| \le 2\sqrt{3h}$. Using the Lipschitz bound for φ_h (A.48) and the lower bound (A.49),

$$|\mathbb{E}[\varphi_h(Y_0)] - \varphi_h(0)| \le \mathbb{E}|\varphi_h(Y_0) - \varphi_h(0)| \le C\sqrt{h} \le C\sqrt{h}\,\varphi_h(x).$$

Apply Lemma 4.19,

$$|P_0\varphi_h(x) - \tilde{P}_0\varphi_h(x)| \le Ch\varphi_h(x), \qquad x \in [0, \sqrt{3h}].$$

We now estimate $P_0\varphi_h(x) - \varphi_h(x)$ by the case-by-case calculation in Lemma 3.5.

(1) For $x \in [0, \sqrt{3h}/2]$,

$$P_0\varphi_h(x) \le \varphi_h(x)\left(1 - f_h(x) + C_3 h\right),$$

where

$$f_h(x) := \frac{s}{4} \cdot \frac{\tilde{\lambda}(x)}{2}\sqrt{3h} + s \cdot \frac{\tilde{\lambda}(x)x^2}{2\sqrt{3h}}.$$

The proof of Lemma 3.5 shows that, for $s > 16$,

$$f_h(x) \ge x, \qquad x \in [0, \sqrt{3h}/2].$$

(2) Likewise, for $x \in [\sqrt{3h}/2, \sqrt{3h}]$,

$$P_0\varphi_h(x) \le \varphi_h(x)\left(1 - f_h(x) + C_3 h\right),$$

where

$$f_h(x) := \frac{s\tilde{\lambda}(x)}{16}\sqrt{3h} + \frac{s\tilde{\lambda}(x)x^2}{4\sqrt{3h}} + \frac{s\tilde{\lambda}(x)x}{4} + \frac{s\sqrt{3h}}{2} - sx.$$

Again, the proof of Lemma 3.5 shows that, for $s > 16$,

$$f_h(x) \ge x, \qquad x \in [\sqrt{3h}/2, \sqrt{3h}].$$

(3) The preceding SBM comparison covers $[0, \sqrt{3h}]$. The enlarged set B_h contains one additional strip of width $O(h)$, namely $[\sqrt{3h}, \sqrt{3h} + K_B h]$. This strip appears only because the diffusion boundary layer is not exactly the SBM boundary layer, so it must be checked directly. It remains to treat the thin strip $[\sqrt{3h}, \sqrt{3h} + K_B h]$. On this interval P_0 performs standard EM jump without reflection because $x \ge \sqrt{3h}$. Also $x + Z \in [0, R]$ for h sufficiently small, so w_h agrees with the original function used in Lemma 3.5. A direct calculation gives, uniformly for $x \in [\sqrt{3h}, \sqrt{3h} + K_B h]$,

$$\begin{aligned} P_0\varphi_h(x) - \varphi_h(x) &= \left(\frac{3}{4} - \frac{x}{2\sqrt{3h}}\right)e^M + \frac{e^M}{2\sqrt{3h}}\int_{\sqrt{3h}/2}^{\sqrt{3h}+x} e^{-s(z-\sqrt{3h}/2)}\,dz - e^{w_h(x)} \\ &= -e^{w_h(x)}\left[\frac{s}{4}\left(\frac{x^2}{\sqrt{3h}} - 3x + \frac{9}{4}\sqrt{3h}\right) + \mathcal{O}(h)\right]. \end{aligned} \tag{A.56}$$

Thus on this thin strip define

$$f_h(x) := \frac{s}{4}\left(\frac{x^2}{\sqrt{3h}} - 3x + \frac{9}{4}\sqrt{3h}\right).$$

For $x \in [\sqrt{3h}, \sqrt{3h} + K_B h]$,

$$f_h'(x) = \frac{s}{4}\left(\frac{2x}{\sqrt{3h}} - 3\right) \le -\frac{s}{8}$$

after decreasing δ_1 so that $2K_Bh/\sqrt{3h} \le 1/2$. Hence f_h is decreasing on the thin strip. Therefore

$$\begin{aligned} f_h(x) &\ge f_h(\sqrt{3h} + K_Bh) \\ &= \frac{s}{16}\sqrt{3h} - \frac{sK_B}{4}h + \frac{sK_B^2}{4}\frac{h^2}{\sqrt{3h}}. \end{aligned} \tag{A.57}$$

Since $s > 16$, after decreasing δ_1 if necessary,

$$f_h(\sqrt{3h} + K_Bh) > \sqrt{3h} + K_Bh.$$

Hence

$$f_h(x) \ge x, \qquad x \in [\sqrt{3h}, \sqrt{3h} + K_Bh].$$

Combining the estimates for $P_0\varphi_h(x) - \varphi_h(x)$ on B_h with (A.47), we obtain

$$P\varphi_h(x) \le \varphi_h(x)(1 - f_h(x) + Ch), \qquad x \in B_h.$$

We now prove the corresponding estimate outside B_h. For $x \notin B_h$, set

$$f_h(x) := 0.$$

Since $\Omega_{br} \subset B_h$, if $x \notin B_h$, then the true scheme is outside the boundary layer and the standard Euler–Maruyama update is nonnegative for every possible Z. Thus

$$X_{k+1} = x + \sigma(x)Z + b(x)h.$$

By Assumption 4, there exists $L > 0$ such that

$$|b(x)| + |\sigma(x)| \le L(1 + x), \qquad x \ge 0.$$

If we can assume global boundedness on drift and diffusion, this part is straightforward because the usual Taylor expansion gives an order-h one-step bound for the test function under bounded coefficients. Without global boundedness assumption, we analyze $P\varphi_h(x)$ in

$$B_h^c \cap [0, R/2], \qquad [R/2, 4R], \qquad [4R, \infty).$$

The first region exploits the piecewise linear structure of $w(x)$ and boundedness of $\sigma(x)$, $b(x)$ in the compact domain. The second region exploits the smoothness of φ and the boundedness of drift and diffusion. The third region, φ is constant by construction and we can make sure the next step is within the constant region by linear growth assumption.

(1) Consider $x \in B_h^c \cap [0, R/2]$. On $[0, R]$, b and σ are bounded. For h sufficiently small, every possible value of $x + \sigma(x)Z + b(x)h$ lies in $[0, R]$. Since $x \notin B_h$, we also have $x > \sqrt{3h}/2$. On $[0, R]$, the function w_h is flat-linear, and for $y \in [0, R]$ and $x > \sqrt{3h}/2$,

$$w_h(y) - w_h(x) \le -s(y - x).$$

Therefore, by Taylor expansion around 0, using first and second moment of uniform random variable and the boundedness of σ, b in $[0, R]$, there exists C_4 is independent of x and h such that

$$\frac{P\varphi_h(x)}{\varphi_h(x)} \le \mathbb{E}\left[e^{-s(\sigma(x)Z + b(x)h)}\right] \le 1 + C_4h.$$

(2) Consider $x \in [R/2, 4R]$. For h sufficiently small, every possible value of

$$Y := x + \sigma(x)Z + b(x)h$$

lies in $[R/4, 5R]$. Also $\sqrt{3h}/2 < R/4$, so φ_h is C^2 on $[R/4, 5R]$. There exist constants $c_\varphi, C_\varphi > 0$, depending on s, M, R and χ but not on x or h, such that

$$0 < c_\varphi \le \varphi_h(y), \qquad |\varphi_h'(y)| + |\varphi_h''(y)| \le C_\varphi, \qquad y \in [R/4, 5R].$$

Taylor expansion gives

$$\varphi_h(Y) = \varphi_h(x) + \varphi_h'(x)(Y - x) + \frac{1}{2}\varphi_h''(\xi)(Y - x)^2$$

for some ξ between x and Y. Taking expectations and using $\mathbb{E}Z = 0$ and $\mathbb{E}Z^2 = h$,

$$|P\varphi_h(x) - \varphi_h(x)| \le Ch.$$

Since $\varphi_h(x) \ge c_\varphi$ on this compact interval, this implies

$$P\varphi_h(x) \le \varphi_h(x)(1 + C_5 h), \qquad x \in [R/2, 4R],$$

where C_5 is independent of x and h.

(3) Consider $x \ge 4R$. By the linear-growth assumption, for all possible Z,

$$x + \sigma(x)Z + b(x)h \ge x - L(1 + x)(\sqrt{3h} + h).$$

Since $R \ge 1$, $1 + x \le \frac{5}{4}x$ for $x \ge 4R$. After decreasing δ_1 so that

$$\frac{5}{4}L(\sqrt{3h} + h) \le \frac{1}{2},$$

we get

$$x + \sigma(x)Z + b(x)h \ge \frac{x}{2} \ge 2R.$$

Thus both x and X_{k+1} lie in the region where w_h is constant. Therefore

$$P\varphi_h(x) = \varphi_h(x), \qquad x \ge 4R.$$

Taking C_1 to be larger than the constants appearing in the estimates above, and decreasing δ_1 if necessary, we obtain (A.45) for all $x \ge 0$ and all $h < \delta_1$. Moreover, by construction,

$$f_h(x) \ge x\mathbf{1}_{B_h}(x).$$

Since f_h is supported on B_h and the formulas above are all of order $\sqrt{h}$, there exists $C_6 > 0$, independent of x and h, such that

$$0 \le f_h(x) \le C_6\sqrt{h}, \qquad \forall x \ge 0, \quad h < \delta_1. \tag{A.58}$$

We now introduce the time-dependent supersolution

$$V(t_i, x) := e^{K_T(T - t_i)}\varphi_h(x), \qquad i = 0, \dots, N. \tag{A.59}$$

The constant $K_T > 0$ will be chosen below. Using (A.45),

$$\begin{aligned} PV(t_i,x) - V(t_i,x) &= e^{K_T(T-t_i-h)} P\varphi_h(x) - e^{K_T(T-t_i)}\varphi_h(x) \\ &\le e^{K_T(T-t_i)}\varphi_h(x)\left[e^{-K_T h}(1 - f_h(x) + C_1 h) - 1\right]. \end{aligned} \tag{A.60}$$

After decreasing δ_1 if necessary,

$$e^{-K_T h} = 1 - K_T h + r_{K_T}(h), \qquad |r_{K_T}(h)| \le C_7(K_T)h^2.$$

Using this expansion in (A.60), together with (A.58), gives

$$PV(t_i,x) - V(t_i,x) \le -e^{K_T(T-t_i)}\varphi_h(x)\left(f_h(x) + (K_T - C_1)h - C_8(K_T)h^{3/2}\right).$$

Choose

$$K_T := C_1 + 1.$$

After decreasing δ_1 again, we may assume

$$(K_T - C_1)h - C_8(K_T)h^{3/2} \ge 0.$$

Hence

$$PV(t_i,x) - V(t_i,x) \le -e^{K_T(T-t_i)}\varphi_h(x) f_h(x). \tag{A.61}$$

Finally, decrease δ_1 once more so that

$$M - s\left(\frac{\sqrt{3h}}{2} + K_B h\right) \ge 0.$$

Then $w_h(x) \ge 0$ on B_h, so $e^{K_T(T-t_i)}\varphi_h(x) \ge 1$ on B_h. Combining this with (A.46) and (A.61), we conclude that

$$PV(t_i,x) - V(t_i,x) \le -x\mathbf{1}_{B_h}(x), \qquad i = 0,\dots,N-1, \quad x \ge 0. \tag{A.62}$$

Define

$$U_B(t_i,x) := \mathbb{E}\left(\sum_{k=i}^{N-1} X_k \mathbf{1}_{B_h}(X_k) \,\middle|\, X_i = x\right).$$

By Lemma 3.9, applied with $g(t_i,x) = x\mathbf{1}_{B_h}(x)$, we have

$$PU_B(t_i,x) - U_B(t_i,x) = -x\mathbf{1}_{B_h}(x), \qquad i = 0,\dots,N-1,$$

with terminal condition $U_B(T,x) = 0$. On the other hand, $V(T,x) = \varphi_h(x) \ge 0 = U_B(T,x)$, and (A.62) shows that V is a supersolution. A backward induction, exactly as in the proof of Lemma 3.5, gives

$$U_B(t_i,x) \le V(t_i,x), \qquad i = 0,\dots,N, \quad x \ge 0.$$

Since $w_h(x) \le M$ for all $x \ge 0$,

$$V(t_i,x) \le e^{K_T T + M}, \qquad i = 0,\dots,N, \quad x \ge 0.$$

Therefore

$$U_B(t_0,x) \le e^{K_T T + M}, \qquad x \ge 0.$$

Using $\Omega_{br} \subset B_h$, we get

$$\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \mathbf{1}_{\Omega_{br}}(X_k) \,\middle|\, X_0 = x\right) \le U_B(t_0, x) \le e^{K_T T + M}.$$

If X_0 is random, taking expectation and using the tower property gives

$$\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \mathbf{1}_{\Omega_{br}}(X_k)\right) \le e^{K_T T + M}.$$

Thus the desired bound holds with $C = e^{K_T T + M}$, which is independent of h. $\square$

Proof of Lemma 4.5. It suffices to prove the one-step estimate

$$\mathbb{E}_x(X_{k+1}^4) \le (1 + Ch)x^4 + Ch,$$

for a constant C independent of x and h.

First consider $x \in \Omega_e$. In this region the scheme makes a standard Euler–Maruyama jump, and the absolute value is inactive, so

$$X_{k+1} = x + b(x)h + \sigma(x)Z_{k+1}, \qquad Z_{k+1} \sim \mathrm{Unif}[-\sqrt{3h}, \sqrt{3h}].$$

Using $\mathbb{E}Z_{k+1} = 0$, $\mathbb{E}Z_{k+1}^3 = 0$, $\mathbb{E}Z_{k+1}^2 = h$, and $\mathbb{E}Z_{k+1}^4 = \frac{9}{5}h^2$, we have

$$\begin{aligned}
\mathbb{E}_x(X_{k+1}^4) &= (x + b(x)h)^4 + 6(x + b(x)h)^2\sigma^2(x)h + \frac{9}{5}\sigma^4(x)h^2 \\
&= x^4 + 4x^3 b(x)h + 6x^2 b^2(x)h^2 + 4xb^3(x)h^3 + b^4(x)h^4 \\
&\quad + 6x^2\sigma^2(x)h + 12xb(x)\sigma^2(x)h^2 + 6b^2(x)\sigma^2(x)h^3 + \frac{9}{5}\sigma^4(x)h^2.
\end{aligned} \tag{A.63}$$

By the linear-growth assumption, $|b(x)| + |\sigma(x)| \le L(1 + x)$. Hence, using $h < 1$ and $(1 + x)^4 \le C(1 + x^4)$ for $x \ge 0$, each term after x^4 is bounded by $Ch(1 + x^4)$. For example,

$$|x^3 b(x)h| \le Chx^3(1 + x) \le 2Ch(1 + x^4),$$

$$x^2 b^2(x)h^2 + x^2\sigma^2(x)h \le Chx^2(1 + x)^2 \le 3Ch(1 + x^4),$$

and the remaining terms are bounded similarly by $Ch(1 + x)^4 \le Ch(1 + x^4)$. Therefore,

$$\mathbb{E}_x(X_{k+1}^4) \le x^4 + Ch(1 + x^4) \le (1 + Ch)x^4 + Ch, \qquad x \in \Omega_e.$$

Next consider $x \in \Omega_{br}$. By Lemma 4.1, after decreasing δ if necessary, $x \le C\sqrt{h}$ and b, σ are bounded on the boundary layer. The scheme either jumps to 0 or makes a reflected Euler–Maruyama-type update, so in either case

$$X_{k+1} \le x + \sigma_2\sqrt{3h} + B_M h \le C\sqrt{h}.$$

Hence

$$\mathbb{E}_x(X_{k+1}^4) \le Ch^2 \le Ch \le (1 + Ch)x^4 + Ch, \qquad x \in \Omega_{br}.$$

Combining the two cases gives

$$\mathbb{E}_x(X_{k+1}^4) \le (1 + Ch)x^4 + Ch, \qquad x \ge 0.$$

Taking expectation with respect to X_k yields

$$\mathbb{E}(X_{k+1}^4) \leq (1 + Ch)\mathbb{E}(X_k^4) + Ch.$$

By the discrete Gronwall inequality, for $0 \leq k \leq N$ and $Nh = T$,

$$\mathbb{E}(X_k^4) \leq e^{CT} \left(\mathbb{E}(X_0^4) + CT\right).$$

Therefore,

$$\sup_{0 \leq k \leq N} \mathbb{E}(1 + X_k^4) \leq C_T,$$

where C_T is independent of h. $\square$